The Cybernetic Economic Model

Draft of an Economy that Regulates Itself

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About this draft

This is a working draft. We are publishing it while it is still taking shape, so that it can be read, tested and contested before it is finished.

Written: Part I and Part II (Chapters 1–11), and from Part III Chapters 12 to 16 and the case study in Chapter 20.

Not yet written: Chapters 17 to 19 (levels, AI, who regulates the regulator), Chapter 21 (the transition), Part IV (what would refute this design, conclusion) and the appendix. The outline that follows shows what they are meant to contain.

Deliberately open: the setpoints. This book does not decide which direction an economy should take; that has to be found democratically. It designs an economy that can follow any such direction and change course without breaking.

Every version carries a number and a date, so that every comment can be matched to the text it refers to. A German version with the same number follows each English one.

Outline

Guiding idea

Title: **The Cybernetic Economic Model.** Subtitle: *Draft of an Economy that Regulates Itself.* The book stands on its own, outside the Megamachine series.

The subtitle draws a deliberate line. The cybernetic experiments of the twentieth century, in the GDR as in Chile, used control loops to make a planned economy possible. This book aims at a cybernetic, market-based overall system: the freedom to decide stays with people and enterprises, while the rules and the design make sure the whole remains stable. That includes positive feedback where it makes growth and the new possible, and the right transition from positive to negative feedback.

Today's economy is dominated by positive feedback loops which, without human corrections, tend towards excess, bubbles and collapse. The book drafts an order with balancing feedback built in from the start, instead of demanding it after the fact. It moves from describing to designing.

Which direction humanity wants to take is not for this book to decide; that is a subject of its own and belongs to the people. The book drafts an economy that can follow any democratically chosen direction and change course without breaking. Only two limits are fixed: ecological carrying capacity as a fixed condition, and no system that steers the people it is meant to serve.

Part I: The Economy as a Control Loop

The diagnosis starts with a false opposition. A truly free market is not the opposite of order. It needs structures and rules that secure its lasting operation without catastrophes and without constant

human intervention. The prevailing model leaves out precisely these rules, and so it describes self-reinforcement instead of self-regulation.

1. **Market or plan: a false opposition.** The planned economy failed for lack of feedback; the actually existing market fails for lack of negative feedback. A free market needs rules that work by themselves, just as a football match becomes free through its rules, not through their absence. Link to the Freiburg School around Walter Eucken, and why rules that are merely laid down are not enough if they do not themselves regulate.
2. **What a control loop is.** Images first: thermostat, cyclist, body temperature, centrifugal governor. Negative and positive feedback.
3. **The textbook model drawn as a control loop.** Environment, state, technology, preferences and power count as "given from outside". Without them there is no balancing.
4. **The positive feedback loops of the present.** Wealth becomes influence, influence becomes legislation. The cycle of private credit funds. The ratchet that prevents retrofitting.
5. **Where price fails.** The seven fields of failure of price formation.

Part II: Four Lines

The map: four schools of thought have so far stood side by side, unconnected. This book brings them together in one place. A contrast study comes first: what became of cybernetics where it was used from the centre.

6. **Cybernetics from the Centre.** How the Soviet Union, the GDR, Chile and China used cybernetics to steer from the top, and why they failed wherever the feedback was switched off.
7. **Stafford Beer.** The structure of viable systems, Cybersyn in Chile and why it failed.
8. **Frederic Vester.** The biocybernetic basic rules as fixed limits.
9. **Cockshott and Cottrell.** Calculating without a market, and the problem of calculating in labour time alone.
10. **Tiqqun and Overwijk.** The warning against the system that steers people.
11. **What holds and what does not.** The synthesis: structure from Beer, limits from Vester, method of calculation from Cockshott, warning from Tiqqun.

Part III: The Design

The core of the book: negative feedback is built, not preached.

12. **The basic principle.** Build negative feedback in instead of demanding it; move reactions out of the moment and into a structure built beforehand, leniency as a component.
13. **The switchover point.** When positive feedback is right and when it must turn into negative feedback. Three ways: an end built in from the start (as with a patent), a threshold on a measurable quantity (like the S-curve in nature), a public decision taken in advance rather than after the event. The counter-check: infant-industry tariffs whose end was decided by the beneficiaries themselves.

14. **Forms of ownership in the control loop.** All forms of enterprise on the same test bench, with their strengths and their failures. Form follows function.
15. **The kinds of money.** Kinds of equal value that differ only in their purpose: consumption money, resource-bound consumption money, investment money, subsidy money, perhaps democracy money.
16. **Signals beyond price.** Ecological limits as a fixed condition. Feedback as a duty of every system.
17. **Levels.** What is regulated in the municipality, what in the region, what in Europe.
18. **AI as a sober component.** What it can do and what it cannot.
19. **Who regulates the regulator?** Democratic control based on the three blueprints: representatives without party, Mandate without Place with subject mandate, Transparency Council. Do not limit how long offices are held; decouple office from self-preservation instead. The sensor sits outside the institution being controlled.
20. **A case study.** The inventors' cooperative.
21. **The transition.** Pilot zones, sequence, resistance.

Part IV: The Test

22. **What would refute this design.** The strongest objections, argued out in the open.
23. **Conclusion.**

Appendix: Spatial models. The tetrahedron for the magic square, the double pyramid of the factors of production with the diagonals capital – land and labour – knowledge, the double pyramid of innovation. Poles and rings as the book's shared visual language.

Building blocks and open questions

Diagnostic material, mechanics and a case study are in hand; the main open item is the kinds of money.

Chapter	Existing building block
3, 5	The neoclassical model as a control loop (April 2026); “Homo Economicus” (May 2026): a view of humanity that measures only one quantity
4	Hysteresis paper, cycle of private credit funds; “The Cruise Control”; “The Carousel”
6	“Cybernetics from the Centre” (October 2026), contrast study on the Soviet Union, the GDR, Chile and China
7–11	Research on the four lines (May 2026); “Paradigms – the Invisible Walls” for the synthesis
12	“Retribution as Architecture”, “The Construction of Mercy”, “The Cruise Control” (incorporated)
13	“The Carousel” (incorporated)
14	“ZONA NOVA”: cooperative ownership on the Mondragón model
15	“Money – Tool or God?”: credit money, interest paradox, compulsion to grow, jubilee year
16	“The Last Step” as an illustration on a small scale
18	“ZONA NOVA”: AI administration with a citizens’ council veto
19	POLIS, “Mandate without Place”; “ZONA NOVA”: citizens’ council
20	Paper on the inventors’ cooperative

Chapter	Existing building block
21	“ZONA NOVA” as a pilot zone; “When Paradigms Fall”
22	“Paradigms – the Invisible Walls”: Kuhn’s anomalies as a test standard
Appendix	Three spatial models

- ☐ Write out the seven fields of failure of price formation
- ☐ Kinds of money: final catalogue of kinds and bookkeeping within the enterprise
- ☐ Calculating in labour time: what takes its place
- ☐ Keep Chapter 15 (kinds of money) open enough for a co-author from business administration to join

Chapter 1: Market or Plan – a False Opposition

The old quarrel

The twentieth century thought of the economy as an either-or. On one side stood the market: millions of people decide for themselves what to buy and what to make, and prices turn these millions of decisions into an order. On the other side stood the plan: a central authority lays down what is produced, how much, and for whom.

The quarrel was not settled by argument but by walls and tanks, and in the end by the collapse of one side. When the Wall fell in 1989, the question seemed answered. The plan had lost, so the market had won. In the same year Francis Fukuyama spoke of the end of history.

That conclusion was hasty. One answer being wrong does not make the other one right. Thirty years later the winner has lived

through its own collapses: the financial crisis of 2008, the euro crisis, the selling-off of the fishing grounds, the climate crisis. Each time, states stepped in that by the pure doctrine should never have intervened at all. A system that has to be rescued in every major crisis has not won the quarrel. It has merely held out longer.

This chapter argues that the opposition between market and plan is falsely posed. Both are attempts to steer an economy. Both fail, if at different points, for the same reason: they lack a property that every lastingly stable system has.

Why the plan failed

In the Soviet Union people told a joke about a nail factory. When its plan was set in pieces, it produced millions of tiny nails that nobody could use. When it was set in tonnes, it produced a few gigantic nails, just as useless. Either way, the plan was fulfilled.

The joke goes to the heart of it. The factory reported upwards what the authority counted, not what people needed. The joiner who could not find the right nails had no way of telling the factory. His shortage stayed invisible. Between need and production, the feedback was missing.

The economists Ludwig von Mises and Friedrich August von Hayek described this problem early on. Mises argued in 1920 that without market prices a planning authority could not calculate which use of scarce resources made the most sense. Hayek showed in 1945 that the knowledge required is spread across millions of heads: the baker who knows that tomorrow is market day, the farmer who knows that her field is too wet. No central office can gather this knowledge. Price bundles it without anyone having to gather it.

In the language of cybernetics, the planned economy was a steering system without sufficient feedback. It sent commands downwards, but the effect of those commands arrived at the top distorted or not at all. Such a system cannot correct itself. It can only command until reality will no longer be commanded.

Why the actually existing market fails

The market has what the plan lacked: feedback. When tomatoes become scarce, their price rises. Some people buy fewer, some growers plant more, and the price falls again. That is a control loop, and a balancing one. The higher price counteracts the scarcity that caused it. Cybernetics calls this negative feedback. It is why the tomato market works year after year without anyone directing it.

But the same feedback does not act in the same direction everywhere. With shares, land or cryptocurrencies, a rising price often triggers not less demand but more. Anyone who sees flats getting dearer buys sooner, before they get dearer still. So the price rises further. The loop reinforces itself until it breaks. Cybernetics calls this positive feedback. Every speculative bubble, from the tulip bulbs of seventeenth-century Holland to the American mortgages of 2008, is positive feedback.

And there are areas where no feedback arrives at all. Off Newfoundland, cod was fished for decades. Between 1962 and 1992 the spawning stock shrank by more than nine tenths. The price of fish did not show this in time, because every boat that landed a fish sold it at the market price, no matter how many were still swimming in the sea. In July 1992 Canada imposed a fishing ban. Some thirty thousand people lost their jobs. The market had a signal for the value of the fish caught, but none for the value of the fish not caught.

So the market does not fail because it has no feedback. It fails where its feedback acts in the wrong direction or does not capture the decisive quantity. These places are not exceptions at the margins. They are the financial markets, land and the natural foundations, the very areas on which all the others rest.

The roundabout

What an order looks like that is neither plan nor mere laissez-faire is shown by a structure almost everyone drives past every day.

A crossroads can be ordered in three ways. The first is the traffic light. It corresponds to the plan: a central authority decides who may go when. It works, but it does not know what is happening on the road. At three in the morning you sit at a red light with no other car in sight. The second is the crossroads without a rule, where everyone drives as they please. With little traffic that goes well; with heavy traffic it ends in a crash or a standstill. That corresponds to the market without order.

The third is the roundabout. It consists of an island in the middle and a single rule: whoever is on the roundabout has right of way. Nothing more is needed. Each driver decides for himself when to enter, as freely as at the crossroads without a rule. And yet the traffic orders itself. When the roundabout is full, no one else can get in, and it empties again. The design itself ensures that nobody races straight across the junction at full speed. There is no central authority, no policeman on the corner, no intervention case by case.

The roundabout is free because it is built. Its negative feedback lies neither in the drivers' good intentions nor in an authority, but in the shape of the structure and in a rule so simple that anyone can follow it. That is the pattern this book seeks for the

economy: few rules, a clever design and freedom for everything else.

Eucken and the Freiburg School

The idea that a free market needs a built order is not new. In Germany Walter Eucken put it most clearly. The Freiburg economist, who died in 1950, saw two dangers at once: the planned economy, which he had experienced in the war economy, and the market left to itself, which leads to cartels and monopolies. His answer was the competitive order. The state should not steer the course of the economy, but set the framework within which competition works.

In his *Grundsätze der Wirtschaftspolitik* [Principles of Economic Policy], published after his death in 1952, Eucken lists the building elements of this framework: a stable currency, open markets, private property, freedom of contract, a reliable economic policy and liability. On the last of these he writes the sentence: “Whoever has the benefit must also bear the damage.” On top of this come rules against monopolies. Ludwig Erhard’s social market economy and the German cartel law of 1957 build on this thinking.

Read cybernetically, Eucken’s liability principle is the purest form of negative feedback. It leads the consequences of a decision back to whoever made it. Anyone who does business riskily and loses bears the loss, and that keeps him and others careful in future. How strongly such a return works is shown by German accident insurance of 1884: once employers had to pay for the accidents in their own works, they built guards into their machines.

And yet the Freiburg School stopped at a decisive point. Its rules are set by people and enforced by authorities, and both are exposed to pressure from those the rules are meant to bind.

Eucken knew this. He called for a strong state standing above the interest groups. But he gave no answer to how such a state protects itself against the positive feedback in which wealth becomes influence and influence becomes laws that increase the wealth.

The financial crisis of 2008 showed what happens then. Banks that were too big to fail were rescued with taxpayers' money. Their owners and managers had had the benefit; the taxpayers bore the damage. Eucken's sentence was not abolished. It was suspended at the decisive moment, by people under pressure, for good reasons. A rule that can be suspended in an emergency is no rule in an emergency.

What free means

Whoever says "free market" usually means a market in which nobody intervenes. This idea confuses freedom with the absence of rules. A football match without rules would not be freer; it would be a brawl that the strongest wins. Only the rules make the game free: every player can do what he likes as long as he keeps to them, and precisely because of that a game emerges and not a scrum.

In this sense today's market is not free. It is steered all the time, only not by rules but by interventions: rescue packages, interest-rate decisions, subsidies, exemptions. Each of these interventions comes late, under time pressure and under the influence of whoever shouts loudest. That is the opposite of a roundabout. It is a crossroads without a rule where, in every jam, a policeman turns up and waves people through as he sees fit.

A market is only free when its rules work by themselves: when the consequences of a decision return to whoever makes it, when reinforcing loops hit a limit before they sweep everything along

with them, and when the natural foundations send a signal before they are exhausted. Then no intervention is needed at the moment of crisis, because the crisis does not arise in the first place, or stays small. Without human intervention does not mean without people. People build the order and decide on it. They should simply not have to decide in a moment of panic.

The task

The plan failed because it lacked feedback. The actually existing market fails where it lacks negative feedback. So the question is not who decides, the state or the market. The question is where the feedback loops lie in an economy, in which direction they act, and whether they are built or left to chance.

That is a question for designers, not for confessions of faith. In engineering it was answered long ago. Ever since James Clerk Maxwell calculated the centrifugal governor of the steam engine in 1868, we have known when a regulated system stays stable and when it builds up into oscillation. Our body holds its temperature to within a few tenths of a degree without our thinking about it. A forest regulates water and nutrients over centuries without a central office. Only the economy, the system on which all other social systems depend, do we leave to a mixture of luck and fire brigade.

The following chapters proceed in three steps. Part I shows more precisely what a control loop is and where today's economy is dominated by positive feedback. Part II examines what previous attempts at a cybernetic economy achieved and why they failed. Part III drafts an order with negative feedback built in from the start, as in a roundabout, not as in a traffic light.

If we have learnt to build a crossroads so that it orders itself, why should that be impossible with an economy?

Chapter 2: What a Control Loop Is

The shower

Everyone regulates every day without knowing the word. You step into the shower and the water is too cold. You turn the lever towards warm. Nothing happens, because the warm water takes a few seconds to come through the pipe. You turn it further. Then it arrives, and too hot. You turn it back, too far again, and stand for a moment in cold water. After three or four swings you have found the right temperature.

Those thirty seconds hold almost everything this book has to say about the economy. There is a goal, the pleasant temperature. There is a measurement, the skin. There is an intervention, the lever. And there is a delay between intervention and effect, which turns a simple task into a back-and-forth. Whoever understands the shower understands control loops.

The four parts of a control loop

The thermostat on a radiator shows the parts of a control loop neatly separated.

1. **The setpoint.** You set the dial to twenty degrees. That is the goal.
2. **The sensor.** A measuring element in the thermostat senses how warm the room actually is. That is the measured value.
3. **The controller.** It compares setpoint and measured value. If it is too cold, it gives a signal.
4. **The actuator.** The valve opens and hot water flows into the radiator.

The room itself, into which the heat flows and out of which it escapes again through windows and walls, is called the controlled system in control engineering. The loop closes because the changed room temperature arrives back at the sensor. Hence the name: the effect returns to the cause.

If one of these parts is missing, nothing works. A thermostat without a sensor heats blindly, like the planned economy of the previous chapter. A thermostat without an actuator measures precisely and changes nothing, like a statistics office whose figures nobody reads. And a thermostat without a setpoint does not know what to regulate towards. We will come back to this last case, because in the economy it is the most common.

Negative feedback

With the thermostat, the feedback counteracts the deviation. If it is too cold, the heating comes on; if it is too warm, the valve closes. This kind of feedback is called negative feedback. It keeps a system close to its goal, whatever acts on it from outside.

Nature is full of it. Our body holds its temperature at about 37 degrees, whether we are standing in the snow or sitting in the sauna. If it gets too warm, we sweat; if it gets too cold, we shiver. A cyclist does not fall over, because without thinking he steers in the direction he is about to topple. A predator-prey relationship in the forest settles into balance: when there are many hares, the foxes multiply until they eat so many hares that their own numbers fall again.

Engineering has its textbook case too. In 1788 James Watt fitted a centrifugal governor to his steam engines: two balls on a rotating shaft. If the engine runs too fast, the balls fly outwards and, through a linkage, close the steam valve. If it runs too slowly, they

sink and open it. No one has to stand by the engine. The rule lies in the structure, as with the roundabout.

Negative feedback is the reason anything lasting exists at all. Everything that endures, living things, landscapes, well-built machines, keeps itself in balance through negative feedback.

Positive feedback

Anyone who has been at an event where a microphone stood too close to a loudspeaker knows the opposite. A faint sound is amplified, comes out of the loudspeaker, gets back into the microphone, is amplified again, and within a fraction of a second there is a deafening howl. Here the feedback acts not against the deviation but in its direction. That is positive feedback.

Positive feedback knows only one goal: more of the same, until something breaks. An avalanche starts with a slab of snow and sweeps along more and more snow the bigger it gets. A bank run starts with a rumour. Anyone who sees others withdrawing their money withdraws theirs too, and that is exactly what makes the rumour come true. In September 2007 people in England queued outside the branches of Northern Rock, the first such scene in Britain for more than a hundred years.

Positive feedback is not always harmful. In childbirth, contractions and hormones reinforce each other until the child is born. A fire you light is meant to spread. What matters is that a useful positive feedback has an end built in: the birth ends the contractions, the wood in the stove is burnt up at some point. Positive feedback becomes dangerous where no end is built in, where it only stops when the system itself breaks.

Delay: the pig cycle

Back into the shower. Its problem was not that the feedback acted in the wrong direction. It was negative feedback; it just came too late. Whoever acts on yesterday's effect always overcorrects, and the system swings to and fro.

The economy has a famous example of this to-and-fro. In 1928 the agricultural economist Arthur Hanau described the pig cycle. When pigs are expensive, many farmers raise more piglets. More than a year passes before these are ready for slaughter. Then they all come onto the market at once, the price falls, the farmers raise fewer piglets, and a year later pigs are scarce and expensive again. Hanau found fluctuations that repeated over decades in a rhythm of a few years. Each individual farmer acted sensibly. Together they produced an oscillation that none of them wanted.

The same pattern turns up far beyond the pigsty. When there is a shortage of engineers or teachers, the subject becomes popular. Five years later too many graduates crowd onto the labour market, and the next intakes avoid the subject until there is a shortage again. With factories, ships and power stations, which take years to build, the game repeats itself with larger sums.

From this follows a rule that economic policy almost never observes: the longer the delay, the more gently the intervention must be made. Whoever counter-steers hard when the delay is long creates the next crisis himself. Negative feedback is only as good as its timing.

Who sets the setpoint?

On the thermostat, someone sets twenty degrees. In the economy, rarely does anyone ask who sets the setpoint and what it is. And yet every economic order has one, spoken or not.

In the planned economy it was spoken: the plan figures, tonnes of steel, numbers of nails. Precisely because it was laid down so exactly, it was fulfilled, even when it missed what was needed. In the market economy the setpoint is unspoken, but no less effective. Companies regulate for profit, stock exchanges for rising share prices, governments for growth in gross domestic product. None of these setpoints includes the question of whether the soil stays fertile, whether people are healthy, or whether the climate stays stable.

A control loop regulates with great fidelity towards the setpoint it is given, even if it is the wrong one. A thermostat set to forty degrees heats reliably until you can no longer bear it. That is why the question of the setpoint is not a technical one but a political one. It belongs in the hands of those who have to live with the consequences. How that can be done is one of the tasks of Part III.

Four questions for every economic order

From the shower, the thermostat and the pigsty come four questions. With them, any economic order can be tested, the existing one as well as the one this book drafts.

1. **Is what matters being measured?** Is there a sensor for the fish not caught, the soil not used up, people's health, or only for what is sold?
2. **Who sets the setpoint, and what is it?** Is it spoken and publicly decided, or does it act tacitly?
3. **Does negative feedback predominate?** Where does positive feedback act, and does it have an end built in?
4. **Does the intervention fit the delay?** Is correction gentle where the paths are long, or hard and too late?

The next chapter puts these four questions to the model that today appears in almost every economics textbook. The result will show why this model cannot foresee the crises it lives through.

If every thermostat needs a sensor, a setpoint and a valve, where are they in our economy?

Chapter 3: The Textbook Model as a Control Loop

How the Model Sees Itself

Almost every economics textbook contains the same picture: two lines that cross. One shows how much is supplied as the price rises, the other how much is demanded. Where they intersect lies the equilibrium. If the price strays from it, the market pulls it back.

This picture is a control loop, and a good one. If the price is too high, goods are left unsold and the traders lower it. If it is too low, the goods sell out quickly and they raise it. The model repeats this loop in three markets. In the goods market, price regulates supply and demand. In the labour market, the wage regulates how many people find work. In the capital market, the interest rate regulates how much is saved and how much is invested. Negative feedback everywhere, everywhere an equilibrium to which the system finds its own way back.

Judged by the four questions of the previous chapter, this would be an exemplary system. It measures, it compares, it adjusts. If reality looks like this, there is no need for intervention and no need for any order beyond the market. This chapter tests whether it does.

The Word “Exogenous”

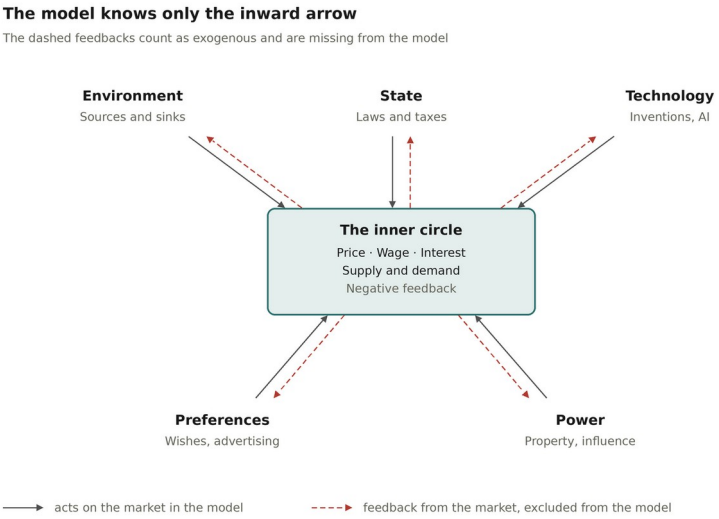
Every model has to leave something out. A map on a scale of one to one would be as large as the country and useless. The question is not whether a model simplifies, but what it leaves out and whether what is left out acts back on it.

Economics has a technical term for what is left out: exogenous, coming from outside. An exogenous variable influences the model but is not influenced by it. It is like the weather for the farmer, something you reckon with but do not change.

In the textbook model, five variables are exogenous: the environment, the state, technology, people’s preferences and the distribution of power. They stand outside the loop. They set the conditions under which price, wage and interest rate seek their equilibrium, but the market does not act back on them.

That would be harmless if it were true. A model of tomato prices may treat sunlight as given, because the tomato market does not change the sun. But with all five variables this feedback exists, and it is not small. By calling them exogenous, the model declares them non-existent by definition.

At a Glance



The model knows only the inward arrow

The model draws only the solid arrows: environment, state, technology, preferences and power act on the market. The dashed arrows, the market acting back on these five variables, do not exist in the model.

Five Feedbacks That Must Not Exist

The environment. The model treats it as a store from which the economy draws and as a rubbish bin that never fills up. In fact every act of production uses up sources and fills sinks: fishing grounds, soils, groundwater, the atmosphere. Once a source is exhausted or a sink is full, this acts back on production with full force, as the cod off Newfoundland has shown. A model that does

not know this feedback sees the collapse only once it has happened.

The state. In the model it sets rules from outside. In fact the economy shapes the state that is supposed to regulate it. Companies and associations finance election campaigns, help write laws, fill seats on advisory boards. If a state cuts taxes on capital, the power of those who lobby for the next cut grows. The rule is not an outer frame but part of the loop.

Technology. In the model it falls from the sky. In fact it arises within the economy, for the economy, and changes the economy. The large language models of artificial intelligence were built with money that data corporations earned from advertising, and they are now changing the very markets in which these corporations earn.

Preferences. In the model everyone knows what they want, and the market serves it. In fact there is an entire industry whose only purpose is to shape preferences: advertising. What people want is partly a product of the same economy that then serves it.

Power. In the model many small sellers meet many small buyers, and none of them can set the price. In fact a few large companies dominate whole markets, and the distribution of ownership is changed by the market itself. Wealth becomes influence, influence becomes laws, laws become new wealth. This is the purest positive feedback in the whole system, and the model treats it as non-existent.

The Four Questions, Put to the Model

Is what matters being measured? What is measured is what has a price. The uncaught fish, the fertility of the soil, an

association's influence on a law have none. They slip past the sensor.

Who sets the setpoint? The model speaks of equilibrium as if it were a neutral goal. But an equilibrium between supply and demand is compatible with any distribution of wealth, including one in which a few own everything. The real setpoint of economic policy is growth, and nobody votes on it.

Does negative feedback prevail? Within the loop, yes. Outside it, in the environment, the state, technology, preferences and power, positive feedback predominates. The model sees only the inner loop and therefore takes the whole to be stable.

Does the intervention match the delay? The model knows no time. All adjustments happen instantly. It cannot represent the pig cycle of the previous chapter, because that arises only when time passes between decision and effect.

The result is clear: the model describes a well-regulated inner loop in the middle of a badly regulated whole. As long as the outer positive feedbacks are small, this goes unnoticed. Today they are large. A model that blanks them out does not describe self-regulation but self-reinforcement, which cannot foresee its own tipping over.

What Economists Know, and Why It Is Not Enough

It would be unfair to claim that economics has overlooked all this. In 1920 Arthur Cecil Pigou described the costs a company imposes on others without paying for them, and proposed taxes to offset them. In 1971 George Stigler showed how the regulated capture their regulators. In 1990 Paul Romer made technology a variable that arises within the economy itself. Since the seventies,

ecological economics around Herman Daly has shown that the economy is part of nature and not the other way round.

The problem is not that this knowledge is missing. The problem is where it sits. It sits in special chapters, footnotes and separate schools, as a correction to a model whose core remains unchanged. Students first learn the inner loop as the rule and then the feedbacks as exceptions. In reality it is the other way round. The feedbacks are the rule, and the undisturbed inner loop is the special case.

An engineer who calculated a bridge and treated the wind as exogenous would not be excused because somewhere there is a textbook on wind loads. He would have to bring the wind into the calculation. This book tries to do the same for the economy: to bring the five excluded variables into the loop and then ask where negative feedback has to be built.

If a model excludes half of its feedbacks by definition: how is it to recognise the crises that come from precisely that half?

Chapter 4: The Positive Feedbacks of the Present

Three Loops, One Direction

The previous chapter showed that the textbook model declares the positive feedback of power non-existent. This chapter shows how it works. It operates on three levels, and on each it turns in the same direction.

On the first level it distributes wealth: whatever has once been appropriated counts as legitimate, and every new gain is

distributed onwards from there. On the second level it circumvents regulation: where one path is blocked, another, less observed, opens up. On the third level it determines which rules can be changed: those that benefit the favoured, easily; the others, hardly at all. Distribution, regulation, rule change. Three loops, and all three reinforce themselves.

The Ratchet of Distribution

A ratchet is the small catch in a ratchet spanner that lets the toothed wheel turn in only one direction. You can crank as much as you like; nothing goes back.

The welfare theory of the textbooks rests on a yardstick that the Italian economist Vilfredo Pareto formulated around 1900. A state is considered efficient if no one can be made better off without someone else being made worse off. In 1954 Kenneth Arrow and Gérard Debreu proved that, under certain conditions, a perfect market reaches such a state. The proof is correct. The question is what the yardstick is worth.

Imagine ten people. One owns 91 parts in a hundred, the other nine one part each. By Pareto's standard this state is efficient, because to give one of the nine something, you would have to take something from the one. The yardstick says nothing about justice, and that was honestly meant. But its effect over time is not neutral.

For when something new comes into being, an invention, a market, a technology, the gain from it initially belongs to no one. Who appropriates it is decided by patents, market power and political influence, that is, by the existing distribution. Once appropriated, it belongs to the new stock, and that again counts as efficient. Every attempt to distribute it differently appears as an

interference with property. Round after round the wheel turns in one direction. That is the ratchet.

The Loop That Relocates Itself

If you block a stream with a dam without sealing off the valley, you have not stopped the stream. The water finds a way round the dam, and it does so where no one is looking.

That is exactly what happened after the financial crisis of 2008. The authorities required banks to hold more equity and to be more cautious with loans. That was a sensible negative feedback. But the demand for credit did not disappear. It migrated to funds that lend without being banks and are therefore not subject to banking rules. By estimates, this market of private credit funds has grown to around 1.8 trillion dollars, in exactly the years in which banks were more strictly regulated.

To secure a steady flow of money, large financial investors bought life insurers. The premiums of millions of policyholders thus flow into the credit funds of the same owners. The policyholders are not owners of these funds but their source of money. Since the beginning of 2026 many investors have wanted their money back, and large funds are limiting withdrawals, some quarter after quarter.

From this follows a sentence that holds for the whole book: negative feedback with gaps does not select against the unwanted behaviour. It selects for the behaviour in the gaps. Every regulation that covers individual institutions and not the design of the whole moves the loop into the next shadow.

The Hysteresis of Rules

Apply a magnetic field to a piece of iron and it becomes magnetic. Switch the field off, and part of the magnetisation remains. The iron remembers. To erase this memory, it is not enough to switch off the old field. You need an opposing field, and it must exceed a certain strength, or nothing happens. Physics calls this memory hysteresis.

Rules in economics and politics behave in the same way. Two examples, more than eighty years apart. On 30 March 2026 the Nasdaq stock exchange changed its rules for the index of its hundred largest stocks: very large companies may now be included just fifteen trading days after their flotation, instead of after three months at the earliest. In June SpaceX went public, at a valuation of 1.77 trillion dollars and with only around five per cent of its shares freely tradable. Index funds had to buy. The rule change benefited the company, the banks and the exchange, and it went through without parliament and without any notable resistance.

The other example is German law on employee inventions. It provides that an employee's invention essentially falls to the employer, and in its basic construction it goes back to a war-economy decree of 1942. It has outlived the regime that created it, and every later reform. The same legal system that rebuilds an index rule in weeks barely moves an inventors' rule in four generations.

The reason is simple, and the economist Mancur Olson described it in 1965. Those who profit from a rule are a small, organised group with a great deal at stake. Those harmed are many, and each individual loses too little to fight back. The rule creates its own defenders: part of the benefits flows into associations, expert reports and campaigns that protect it. The longer it applies, the

more firmly it is anchored. This is not inertia, for inertia would be the same in both directions. It is a ratchet in the toolbox itself.

The Rising Threshold

If an entrenched rule needs an opposing field that exceeds a certain strength: what has that opposing field been in history? The honest answer is: mostly a catastrophe.

The separation of commercial and investment banking in the United States, the Glass-Steagall Act of 1933, came not through insight but through the crash of 1929. Public anger pushed Wall Street's holding force below the threshold for a moment. The law was then dismantled without any catastrophe at all, through decades of patient lobbying, until its most important parts fell in 1999. Built only through collapse, torn down through mere persistence.

On top of that, the threshold rises. The longer a rule applies, the more holding force has accumulated, and the larger the crisis must be that can still break it. The financial crisis of 2008, the biggest shock since 1929, did not break a single fundamental property rule. It moved lending into the shadows and accelerated concentration. In 1982 Olson himself explained the rapid growth of Germany and Japan after 1945 by the fact that defeat had swept away their encrusted interest groups, while victorious Britain suffered from its intact encrustations.

This is the darkest finding of this chapter. A system that can no longer be renewed against its beneficiaries waits for an opposing field that is strong enough. So far these have been collapses and wars. Anyone who does not want that must build the negative feedback before the threshold becomes unreachable.

What Follows from This

The three loops yield three requirements for any order this book designs.

1. **Negative feedback in the design, not in the individual rule.** Whoever regulates individual institutions moves the loop. Only what changes the design of the whole is effective, for instance the question of where a company's surpluses flow and who is liable for its losses. How well each form of enterprise achieves this is examined, by the same measure for all forms, in the chapter on forms of ownership.
2. **Symmetry of changeability.** A rule that gives one group advantages must not be easier to introduce than to abolish. Privileges need a built-in end: they expire after a fixed period unless they are justified and decided anew. Then it is not the abolition that has to find a majority, but the extension.
3. **Build before the threshold rises.** Negative feedback can be introduced with arguments in calm times. In a crisis it is too late, and after the crisis it is often too expensive.

The next chapter asks where exactly price fails as a controller. It is the last stage of the diagnosis before the book examines previous attempts at a different order.

If a ratchet turns only one way: who builds the catch so that it can be released again?

Chapter 5: Where Price Fails

What Price Promises

In Chapter 1, price appeared as what the planned economy lacked: a feedback that bundles millions of scattered pieces of knowledge into a single number. Hayek described it that way in 1945, and he was right. No one needs to know why copper has become scarce. It is enough that its price rises. Then everyone economises on copper, and new mines become worthwhile.

From this idea the textbooks derive four functions. Price signals scarcity. It steers resources to where they are valued most. It rewards whoever produces more cheaply. And it weeds out whoever persistently fails to cover their costs. If all four functions work cleanly, an economy orders itself without a centre.

This chapter does not ask whether price is a good sensor. It often is. It asks at which points it systematically measures wrongly, and whether these points are marginal cases or concern the foundations.

Seven Points at Which It Fails

First: costs borne by others. A price arises where two parties trade. What a third party bears is not in it. Whoever burns coal pays for the coal, but not for the heatwave that destroys a harvest elsewhere decades later. The atmosphere, the groundwater and future generations do not sit at the negotiating table. Economists call these externalities and often treat them as an exception. But almost all production has such effects. They are not the exception but the norm.

Second: unequal knowledge. In 1970 George Akerlof used the used-car market to show what happens when the seller knows

more than the buyer. Because the buyer cannot tell good cars from bad, he pays only a middling price. At that price no one sells a good car, and in the end only the bad ones remain on the market. The same applies to insurance, financial products and medicines. Where knowledge is unequally distributed, price measures not quality but mistrust.

Third: market power. Price regulates only as long as no one can set it. A single search engine today handles around nine out of ten of the world's search queries. Anyone who wants to be found there pays what is asked. With digital platforms there is the added fact that every new user makes the platform more valuable for all the others. That is a positive feedback that leads to monopoly without anyone having to act unfairly.

Fourth: delay. The pig cycle from Chapter 2 shows it: price reports today's scarcity, but the response to it comes only in a year, or in five. Whoever invests according to today's price builds for a scarcity that by then no longer exists. The longer the building takes, with power stations, factories, training, the more strongly the system oscillates.

Fifth: path dependence. Almost all of Europe's railways run on a gauge of 1,435 millimetres because George Stephenson built them that way in England. Whether it was the best no longer matters. Once a network has been built, any change costs more than sticking with it. The same applies to networks of filling stations, power grids, programming languages and cities built for the car. Price compares only the next decision, not the path. It holds on to whatever is already there.

Sixth: the short view. Whoever invests today discounts future returns and damages to their present value. At a discount rate of five per cent a year, damage of a hundred euros that occurs in fifty years is worth less than nine euros today. Economists have

been arguing about which discount rate is right since Nicholas Stern's report on climate change in 2006. The argument shows that what is doing the calculating here is not a law of nature but a decision about how much our grandchildren count.

Seventh: what is not traded has no price. Caring for a child, voluntary work, the pollination of fruit trees by bees, a quiet forest: none of this appears in a price as long as it is not sold. What has no price counts as zero in the calculation. A sensor that reports everything unsold as zero underestimates precisely what everything else lives on.

Read Cybernetically

Each of the seven fields of failure is a specific fault in the control loop. The classification from Chapter 2 makes visible which part is disturbed in each case.

Field of failure	What is disturbed in the control loop	Example
Costs borne by others	The sensor measures only the trade, not the consequences	Heatwave from coal power
Unequal knowledge	The sensor is falsified by one side	Used cars, financial products
Market power	One party holds the actuator fast	Search engine, platforms
Delay	Dead time between signal and effect	Pig cycle, power station build
Path dependence	Hysteresis: the loop remembers the path	Gauge, filling-station

Field of failure	What is disturbed in the control loop	Example
		network
The short view	The setpoint barely counts the future	Climate damage in fifty years
What is not traded	The sensor is blind to this variable	Care, pollination, forest

The table shows that this is not about seven different defects that could be remedied one by one. They are disturbances in every part of the same loop: in the sensor, in the actuator, in the setpoint and in the timing.

One Unit of Account Is Not Enough

One might conclude from this finding that planning was right after all. That would be a false conclusion, and Part Two of this book will examine it closely. The British computer scientist Paul Cockshott has proposed calculating an economy not in money but in labour time. This unit of account, too, is sluggish and open to manipulation, and it is just as blind to soil, water and atmosphere as price is.

Here lies the common error of both camps. Hayek calculates everything in money, Cockshott everything in labour time. Both try to express all the scarcities of an economy in a single number. But there is more than one scarcity. Labour is scarce, and so are energy, land, water, the absorptive capacity of the atmosphere, time and trust. These quantities cannot be converted into a single one without loss.

A doctor who measures only the pulse will miss a poisoning as long as the heart beats regularly. No doctor would abolish the pulse for that reason. He also measures temperature, blood pressure and blood values. An economy likewise needs several sensors working side by side, and several signals that do not vanish into a single number. How this can be done without falling into a new bureaucracy is the question of Part Three. Part of the answer will be that money itself need not be a single kind either.

With this the diagnosis is complete. The market has a good inner loop, but the positive feedbacks of power, the blindness to what is unsold and the short view make the whole tip over. Part Two asks what previous attempts at a cybernetic economy have made of this.

If no doctor is content with the pulse alone: why are we content with price alone in the economy?

Chapter 6: Cybernetics from the Centre

The Soviet Union, the GDR, Chile and China all treated cybernetics as a tool for steering from the top. They did not fail because their computers were too weak. They failed because the feedback was switched off.

The dream of a computer that replaces the market

In 1967 the Polish economist Oskar Lange wrote a sentence that sums up an entire era. The market process, with its cumbersome tâtonnements, appears old-fashioned. “Indeed, it may be considered as a computing device of the preelectronic age.” Were he to rewrite his essay of 1936 today, Lange said, his task would be much simpler: put the simultaneous equations on an electronic computer, and the solution comes in less than a second.

Twenty-two years earlier Friedrich August von Hayek had argued the opposite. The price system, he wrote, is a mechanism for communicating information, “a system of telecommunications”. It enables individual producers to watch merely the movement of a few pointers, as an engineer might watch the hands of a few dials.

Both use the same image, the machine that processes information. They draw opposite conclusions. For Lange the market is a slow computer that a faster one can replace. For Hayek it is a network of sensors that nobody can replace, because the knowledge it reports exists nowhere in one place. The four experiments that follow show who was right, and why the answer is not quite as simple as either camp believes.

Soviet Union: cybernetics against the apparatus

From pseudoscience to miracle weapon. In 1954 the Soviet Short Philosophical Dictionary still called cybernetics “a reactionary pseudoscience”, a tool of imperialist reaction. One year later three scientists rehabilitated it, Sergei Sobolev, Alexei Lyapunov and the military computer scientist Anatoly Kitov, with an article in *Voprosy filosofii*. Within a few years the banned doctrine had become a hope.

Kitov, 1959. In January 1959 Kitov wrote to Khrushchev, proposing to equip large enterprises and agencies with computers and link them into a single management system. The same year came the “Red Book”: a network of military computing centres that would also run the economy. A commission of generals rejected it and demanded punishment. Kitov lost his post, was dismissed from the army and expelled from the Party. Whether the expulsion was permanent or lasted only a year, the sources disagree.

Glushkov's OGAS. From 1962 the Kiev mathematician Viktor Glushkov designed a "Nationwide Automated System" for collecting and processing economic data: one centre in Moscow, up to 200 intermediate computing centres, up to 20,000 local terminals, connected via the telephone network. The plan was to know in real time what happened in every enterprise. On 1 October 1970 the project came before the Politburo. Finance Minister Garbuzov mocked computers that would flash lights and play music in chicken coops. The project was neither approved nor rejected. Before the 24th Party Congress in 1971 it was cut back to isolated solutions. After that every ministry built its own system, and none fitted any other.

The media historian Benjamin Peters drew a bitter conclusion: the first large civilian computer networks emerged among cooperative capitalists, not among competitive socialists. The capitalists behaved like socialists, the socialists like capitalists. OGAS did not fail because of technology, nor because of too much centralism, but because every ministry defended its data as a possession of power.

Kantorovich's shadow prices. As early as 1939 Leonid Kantorovich in Leningrad invented linear programming, prompted by an enquiry from a plywood trust. His method also produced "objectively determined valuations", called shadow prices in the West: numbers that show how scarce a resource is. That was ideologically dangerous, because it looked like marginal utility theory, that is, like bourgeois economics. Open discussion was blocked until 1956. In 1975 Kantorovich received the Nobel Prize, together with Tjalling Koopmans. The irony: precisely the best mathematical theory of planning proved that optimal planning needs scarcity prices.

The reform nobody wanted. In 1962 the economist Evsei Liberman proposed in *Pravda*, under the title “Plan, Profit, Bonus”, that enterprises be measured by profit. The Kosygin reform of 1965 put this partly into practice and cut the number of plan targets from thirty to nine. Then it petered out. The ministries kept issuing contradictory orders, the price reform never came, and from 1969 the old targets returned. After the Prague Spring of 1968 every economic reform was seen as a political risk. In *Building a Ruin* (2024), the historian Yakov Feygin describes how the unresolved struggle between planners and reformers led to stagnation. The escape forward was called the “scientific-technical revolution”: better computers instead of different rules.

GDR: the short decade of cybernetics

In East Germany the curve was even steeper, up and down. In 1961 the philosopher Georg Klaus published *Kybernetik in philosophischer Sicht* (Cybernetics from a Philosophical Perspective). His thesis: cybernetics does not contradict dialectical materialism, it confirms it. At the 6th Party Congress in January 1963 Walter Ulbricht declared that cybernetics was to be particularly promoted. The same year saw the start of the “New Economic System of Planning and Management” (NÖS), designed by Günter Mittag and Erich Apel. Enterprises were to gain more responsibility, profit and prices were to act as “economic levers”. Cybernetics supplied, so to speak, the scientific foundation.

The end came in three steps. On 3 December 1965 Erich Apel, chairman of the State Planning Commission, took his own life on the day the GDR signed a long-term trade agreement with the Soviet Union that he had opposed. After the Prague Spring, Kurt Hager attacked cybernetically minded philosophers for “revisionism” in April 1969. And after Ulbricht’s fall in 1971, Erich

Honecker reportedly declared at the 8th Party Congress that it had now finally been proven that cybernetics and systems research were pseudosciences. This quotation is attested in only one secondary source.

The lesson from the GDR: as long as cybernetics helped the Party improve its planning, it was welcome. As soon as it pointed towards autonomous enterprises, that is, towards genuine control loops without a centre, it became dangerous.

Chile: the exception that is remembered wrongly

Cybersyn, the project of the Allende government from 1971 to 1973, is often seen as the summit of cybernetic central planning. It was meant differently. Chapter 7 describes how Stafford Beer built it: enterprises were to regulate themselves, only a few key figures went to Santiago each day, and an alarm went one level up only if an enterprise had not corrected a deviation itself within a set time. This is not command from above but a nervous system that wakes the centre only when something hurts.

The historian Eden Medina judges both ways: the charge of totalitarian control was exaggerated. But Beer's promise of autonomy was kept only in part. The alarm thresholds were mostly set by engineers, not by workers. For our subject Cybersyn is therefore the most important of the four cases: the only attempt to build cybernetics from the bottom up, and at the same time an example of how quickly the setpoints end up with experts again.

China: systems engineering instead of control loops

The rocket man. In 1954, in the United States, Qian Xuesen wrote the textbook *Engineering Cybernetics*. In the McCarthy era he lost his security clearance and spent years under house arrest. In 1955 he was allowed to leave for China, apparently in exchange

for American prisoners of war. There he built the missile programme and made systems engineering the mindset of the planners. In 1978 he published his programmatic article “The Technology of Organisational Management – Systems Engineering”. Later he developed the theory of “open complex giant systems”, among which he counted society.

Population as a controlled system. Qian’s student Song Jian, a control engineer from missile research, encountered Club of Rome-style population models at a congress in Helsinki in 1978. In 1980 his group calculated an “ideal” population for China of 650 to 700 million and recommended one child per couple. The one-child policy was adopted in September 1980. How large Song’s share was is disputed. The anthropologist Susan Greenhalgh sees him as its real author. Other researchers argue that the Party leadership had already decided in 1979 and used the models only as scientific justification. What is certain: a society was treated like a controlled system, with a setpoint fixed by engineers.

The language of the Party. The term “top-level design” (顶层设计) appeared in 2010 in the proposal for the 12th Five-Year Plan and became central from 2013. In 2020 the Central Committee adopted “adhering to the systems concept” as one of five guiding principles for the 14th Five-Year Plan. The descent from Qian’s systems engineering is recognisable but not spelled out in official texts. We have found no statement by Xi Jinping that explicitly names Qian’s systems engineering as a foundation.

Glushkov with big data. On 19 November 2016 Jack Ma said in Shanghai that over the next thirty years the planned economy would grow ever larger. Thanks to data, one might be able to find the invisible hand of the market. In May 2017 he repeated this in Guiyang and mentioned that economists had dismissed it as nonsense. The same year two scholars at Sichuan University

outlined a “plan-oriented market economy” based on platform data. Other Chinese economists objected in 2020: big data serves macroeconomic regulation but does not replace the market. Logan Wright cites ideological papers that see AI as a means of bypassing market forces. By his figures, the entire AI build-out accounts for less than one per cent of economic output.

No Big Brother cast in one piece. The Social Credit System, adopted in 2014, is usually described in the West as a central score for every citizen. That is wrong. It is mainly a collection of administrative and court data with a focus on companies, little automated and with many local pilot schemes. The “City Brain” in Hangzhou, run by Alibaba since 2016, controls traffic lights and traffic. That is control engineering on a small scale, and there it works.

The measured value lies. The real steering of the Chinese economy runs not through computers but through credit and careers. Economists have shown that provincial leaders were more likely to be promoted the higher their province’s growth, a tournament by key figures. The result is well known. In 2017 Liaoning province admitted that it had inflated its fiscal revenues for 2011 to 2014 by at least twenty per cent. In 2018 Inner Mongolia and the Binhai district of Tianjin corrected their figures, Binhai cutting its 2016 output by a third. As early as 2013 the Party’s Organisation Department had ruled that officials should no longer be judged by GDP alone. The Rhodium Group, where Logan Wright works, estimates actual growth for 2024 and 2025 at two and a half to three per cent; around five per cent was reported.

Positive feedback without an exit. From 2008 to 2015 the debt of the Chinese economy rose from about 150 to about 250 per cent of economic output. Credit drove construction, construction drove

land prices, land sales filled the coffers of local governments, and they received credit again. Including its inputs, the property sector reached about a quarter of economic output. In the catch-up phase this positive feedback was successful. China has not managed the transition to negative feedback. The natural negative feedbacks, bankruptcies, falling prices, unwelcome news, are absorbed by state banks, subsidies and censorship. The error does not disappear. It is exported, as price pressure on other countries' industries.

What follows

Four experiments, one pattern. Everywhere cybernetics was understood as a means of making the centre smart. Nowhere did it fail because the computers were too slow. Four lessons can be drawn.

1. **Whoever is rewarded by key figures falsifies the key figures.** The Soviet plant director reported plan fulfilment, the Chinese provincial leader reports growth. A controller with a falsified measured value regulates into the void, and no computer in the world can help. A self-regulating model needs measured variables that nobody has an interest in falsifying.
2. **Data is power, and power is not shared.** OGAS failed because ministries would not give up their figures. A control loop that runs through an authority is only as strong as that authority's willingness to pass on unwelcome reports.
3. **The centre switches off the negative feedback that disturbs it.** Bankruptcies are prevented, prices fixed, criticism censored. The system looks stable until the pent-up deviation escapes elsewhere: as stagnation in the Soviet Union, as an export flood from China.

4. **Setpoints migrate to the experts.** Even in Beer's Cybersyn, engineers ended up setting the thresholds; in China, missile engineers fixed the number of children. Whoever determines the setpoint governs, however clever the controller.

This yields the distinction the book is aiming at. The Soviet Union, the GDR and China wanted to **steer**: a centre that issues ever more precise instructions with ever better data. A cybernetic economic model in the true sense wants to **regulate**: many control loops that work without a centre, whose feedback cannot be switched off, and whose setpoints are set not by engineers or functionaries but democratically. Hayek was right that prices are sensors. He did not say who ensures that the network is not torn apart by its own positive feedback loops. That is the task of the book.

If the Soviet Union, the GDR and China, with every computer they could get, failed because they switched off the feedback: why should a faster computer succeed where they did not?

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- Kitov's expulsion from the Party: permanent or for one year.
- Wording of the 1954 dictionary entry: transmitted in variants; original text not seen.
- Honecker's "pseudosciences" quotation of 1971: only one secondary source.
- Song Jian's role in the one-child policy: originator or justifier after the fact.
- Number of telex machines in Cybersyn: the often-cited 500 is not confirmed; Medina writes "several hundred".

Chapter 7: Stafford Beer and the Experiment in Chile

A Letter from Santiago

In July 1971 the British management consultant Stafford Beer received a letter from Chile. It had been written by Fernando Flores, a young engineer at the state development agency. Salvador Allende's government had nationalised hundreds of

firms and faced a task at which the Soviet Union had failed: to guide an economy without smothering it with orders. Flores had read Beer's books and asked whether he would like to help.

Beer was no socialist but a successful consultant to British industry. Yet he saw the chance to test his ideas on an entire national economy. For two years he commuted between London and Santiago. The project that came out of it was called Cybersyn. To this day it is the only serious attempt to build cybernetics into a national economy from the bottom up.

This chapter asks what Beer built, what of it held up and what did not. It begins with the model on which everything rested.

The Viable System

Beer asked himself a simple question: what do all systems have in common that survive over long periods even though their surroundings change constantly? His answer was the Viable System Model, worked out in his 1972 book "Brain of the Firm". His template was the human body.

The body consists first of organs that do the actual work: the heart pumps, the lungs breathe, the stomach digests. Each organ largely regulates itself. So that they do not disturb one another, there is coordination: when you run, you breathe faster, and the heart keeps pace. Above this stands an authority that monitors the interplay in the here and now and allocates resources. Beside it, one that looks outwards and ahead: what is coming my way? And at the very top, what the person wants, who they are, what they live for.

Beer translated this into five subsystems that every viable organisation needs.

1. **The operational units.** They do the actual work and regulate themselves as far as possible.
2. **Coordination.** It prevents the units from setting one another oscillating.
3. **Internal steering.** It allocates resources and monitors the interplay. This includes a separate audit channel that occasionally checks directly instead of relying on reports.
4. **The view outwards and ahead.** It observes the surroundings and asks what has to change.
5. **Identity.** It holds the balance between the today of internal steering and the tomorrow of the outward view, and decides where the whole wants to go.

Two ideas make this model special. The first is repetition at every level. Each operational unit is itself a viable system with the same five parts, just as every organ consists of cells that regulate themselves. A factory is such a system, so is the industry it belongs to, and so is the national economy.

The second idea comes from the British cybernetician William Ross Ashby, a companion of Beer's, and is called the law of requisite variety: only variety can absorb variety. A goalkeeper who can only dive to the left will not stop a ball that flies to the right. A headquarters that knows only ten orders for a thousand different situations cannot steer an economy. From this follows Beer's most important lesson: variety must be dealt with where it arises, in the units themselves. Only what cannot be solved below is reported upwards.

Cybersyn

What was built in Chile was modest by today's standards. There were hardly any computers in the country. Beer connected the firms by telex machines – the most frequently cited figure is five

hundred – to a single mainframe in Santiago, first an IBM machine, later a Burroughs. Every day the firms reported a few key figures: production, raw materials, absenteeism. A program called Cyberstride compared these figures with what was to be expected and looked for deviations.

What mattered was what happened next. If the program reported a deviation, the signal went first to the firm itself. It had time to solve the problem. Only if it failed did the report rise one level higher. Beer named these signals after the Greek words for pain and pleasure. They were the counter-design to the planned economy: no orders from above, but alarms from below, and even those only once the lower level was overwhelmed.

In Santiago an operations room was built, designed by a team around the German designer Gui Bonsiepe: seven swivel chairs in a circle, screens with charts on the walls. It looked like the bridge of a spaceship and became the symbol of the project.

Its test came in October 1972. A nationwide strike by lorry owners was meant to paralyse the supply of the capital. Over the telex network the government coordinated the few vehicles that were still running. According to one participant, they managed to supply the city with only about two hundred lorries. By May 1973 around a quarter of the nationalised firms were connected, generating about half the turnover of that sector. Only about twenty factories had been fully modelled.

On 11 September 1973 the military staged a coup. Allende died in the presidential palace. The soldiers destroyed the operations room, and the project was abandoned.

What Cybersyn Showed

Cybersyn was ended by force after two years. It was neither proven nor disproven. And yet it showed three things that matter for this book.

First, that steering without command is possible. The centre in Santiago did not tell the firms what to produce. It only watched and raised the alarm when something got out of hand, and even then only after the firm itself had had the chance to put it right. That is exactly the feedback that the nail factory in Chapter 1 lacked, without taking away its freedom.

Second, that the grace period before the alarm is a form of leniency. A deviation is first of all the business of whoever caused it. Only someone who fails to fix it is reported. This is the same distinction that was drawn in “Retribution as Architecture” between unambiguous and ambiguous triggers: where a problem can be fixed, the opportunity comes before the consequence.

Third, that a feedback system holds up in a crisis. In the strike of October 1972 what mattered was not the size of the computer but that the reports arrived quickly and that someone knew how to interpret them. With very simple means a nervous system came into being where there had been none before.

Where Beer Reaches His Limits

A merciless examination must not stop at admiration. Beer’s model has four limits that have nothing to do with the coup.

It regulates production, not need. Cybersyn knew how much a factory produced. It did not know whether people needed what was produced. The feedback from consumers, which the price supplies in the market, had no sensor of its own in the system.

It does not know nature. The key figures reported concerned raw materials as stock for production, not as a foundation that can be exhausted. That is no reproach to the seventies, but it is a gap that the next chapter must fill.

It does not say who sets the setpoint. The identity of the whole, the fifth subsystem, was in Chile the government. Beer wanted to involve the workers in decisions, but that never got beyond first steps. A model that leaves the question of the setpoint to a single authority is only as good as that authority. In the hands of another government the same telex machines could have served surveillance.

Today it lives mainly in companies. Beer's model is carried on at the University of St. Gallen and in management consulting, for instance by Fredmund Malik. There it helps to organise corporations. The step to the national economy that Chile dared has not been attempted since.

What We Adopt

From Beer this book adopts the structure. An economy with negative feedback needs units that regulate themselves, coordination between them, a level that looks ahead, and one that decides where the whole wants to go. It needs this structure at every level, from the workshop to Europe. It solves problems where they arise and reports upwards only what cannot be solved below, with a grace period for its own correction.

What is not adopted is the operations room as a place where a government sets the setpoint. Beer does not answer this question, and it is the most dangerous one. The other two gaps, the missing feedback from consumers and the missing nature, are filled by the next chapters: nature with Vester, the calculation with

Cockshott, and the danger of surveillance with the critics of cybernetics.

If five hundred telex machines could give a country a nervous system: what could a country build with today's means, and who keeps a hand on the switch?

Chapter 8: Frederic Vester and the Rules of Nature

A Biochemist Sees Systems

Stafford Beer asked how organisations survive. Frederic Vester put the same question to nature. The biochemist, born in 1925 and died in 2003, spent his life trying to understand why a forest, a lake or a body remains stable over long periods, while human systems so often tip over.

His books reached a large public, from "Das Überlebensprogramm" in 1972 and "Neuland des Denkens" in 1980 to "Die Kunst vernetzt zu denken" of 1999, a report to the Club of Rome. His approach was practical. He wanted not only to describe but to provide tools with which a municipality, a company or a transport ministry can recognise its own webs of cause and effect.

What Beer achieves for structure, Vester achieves for limits. He does not say how a system must be organised, but which rules it must not break if it wants to survive.

Eight Rules from Nature

Vester summed up his observations in eight biocybernetic basic rules. Each has a model in nature, and each can be applied to the economy.

1. **Negative feedback must dominate over positive feedback.** Positive feedback may exist, as a driving force, for growth in a phase. But in the long run the balancing loops must be stronger. This is the basic rule on which this whole book stands.
2. **The functioning of the system must be independent of quantitative growth.** A tree stops growing taller at some point and still lives on for centuries. An economy that only works as long as it grows breaks this rule. This is the setpoint that Chapter 2 asked about.
3. **The system must work by functions, not by products.** A person needs not a drill but a hole in the wall, and not a car but a way to work. Whoever calculates by products sells more drills. Whoever calculates by functions asks how the hole can be made most simply.
4. **Use existing forces instead of fighting them.** Vester called this the jiu-jitsu principle, as opposed to the boxer's method. A river that bursts its banks is not mastered with ever higher dykes, but with floodplains in which it is allowed to spread out.
5. **Multiple use.** A roof can drain rain, generate electricity and provide shade. A school building can be an adult education centre in the evening. Whatever serves several purposes needs less material.
6. **Cycles instead of waste.** A forest knows no rubbish. What falls becomes food for something else.

7. **Symbiosis.** Different things make use of one another. A lichen is a fungus and an alga living together where neither could survive alone. In Kalundborg in Denmark, firms have for decades been using one another's waste heat, waste water and by-products.
8. **Planning with feedback.** Products, processes and organisations are designed so that they learn from their effects, not according to a rigid plan.

It is striking how well the rules fit the diagnosis of Part One. The positive feedback loops of power break the first rule, the compulsion to grow the second, the blindness to what goes unsold the sixth. Vester anticipated in the language of biology what control engineering finds in the economy.

The Paper Computer

Vester's best-known tool is astonishingly simple. He called it the Paper Computer, and all it needs is a sheet with rows and columns. In both you write the same variables of a system, for a town, say: traffic, rents, jobs, air quality, green spaces. Then you ask for every pair: how strongly does this variable act on that one? Not at all, weakly, moderately or strongly.

If at the end you add up how strongly each variable acts on others and how strongly it is itself influenced, the variables sort themselves into four groups.

- **Active variables** act strongly and are little influenced. They are the levers. Whoever turns them changes a great deal.
- **Passive variables** are strongly influenced and act little. They are indicators, like a fever for the doctor. Whoever turns them is treating the thermometer.

- **Critical variables** act strongly and are strongly influenced. They are the accelerators. A small push can make the system swing ever higher.
- **Buffering variables** act little and are little influenced. They keep the system calm, but you can move little with them.

The benefit is obvious. Many political measures turn passive variables, because these are visible: unemployment figures, growth rates, poll ratings. The Paper Computer forces you to look for the levers before you act. Its weakness is just as plain. The ratings rest on the judgement of those involved. Whoever seats only the representatives of one side at the table gets their world view back as a matrix.

What Vester Provides and Where He Stops

Vester provides what Beer lacks: nature as part of the system and not as a stockpile at the edge. His first and second rules are the sharpest formulation of what an economic order must achieve. And his Paper Computer is a tool with which the five excluded variables of Chapter 3 can be brought back into the loop.

But Vester stops at three points.

He is designed to be politically neutral. His rules say what marks a healthy system, not who prevents it. The positive feedback of power from Chapter 4 does not appear in his rules. A forest has no lobby. An economy has many.

He has no unit of account. Vester says that cycles are better than waste and functions better than products. He does not say how an economy translates this into its daily decisions, into prices, wages, loans. Between the rule and the till of a craft business the connecting link is missing.

He stayed small. His tools were applied in municipalities, regions and companies, never to an economic constitution. That is no failure, but a gap he left open.

What We Adopt

From Vester this book adopts the eight rules as touchstones. Every proposal in Part Three must be measured against them: does negative feedback dominate? Does it work without the compulsion to grow? Does it calculate in functions? Does it close cycles? Where it breaks a rule, it must say why.

Also adopted is the Paper Computer as a method for telling levers from indicators. In Part Three it is to be applied to the economy itself, with participants from more than one camp at the table.

What Vester leaves open, the unit of account, is the question of the next chapter. There stands a man who believed he had found it.

If a forest has lived for centuries without growth and without waste: which rule of our economy forbids it to learn the same?

Chapter 9: Cockshott, Cottrell and Labour Time

Calculating Without a Market

In 1920 Mises had claimed that without market prices nobody could calculate which use of scarce resources was best. The planned economies of the twentieth century seemed to prove him right. But they calculated with paper, telephones and slide rules. What if one could calculate with computers?

This question was put by the computer scientist Paul Cockshott from Glasgow and the economist Allin Cottrell in their 1993 book “Towards a New Socialism”. Their answer was bold: the computational task of planning could be solved with modern technology. The Soviet Union’s problem had not been planning, but its tools.

To this day the book is the most serious attempt to calculate the economy without a market. That is why it deserves close examination.

The Proposal of Cockshott and Cottrell

The core of the proposal is a unit of account: labour time. Every product has a value that follows from how many hours of work are contained in it, including the hours contained in machines and intermediate products. A loaf of bread containing twenty minutes of work in all costs twenty minutes.

People are paid not in money but in labour credits. Whoever works an hour receives a credit for one hour, less a share for communal tasks. With these credits they buy goods. The credits are cancelled on purchase; they do not circulate and cannot be lent. So whoever holds credits cannot make them work. Capital in today’s sense cannot arise.

The idea is old. As early as 1832 Robert Owen founded an exchange in London where goods were traded for labour notes. What is new with Cockshott and Cottrell is the computer. With input-output tables, which show how much each industry draws from every other, the labour content of all products can be calculated and a plan drawn up that matches need and production. The authors show that this computational effort does not explode as the economy grows, but remains manageable.

Politically they propose a democracy on the Athenian model: bodies whose members are chosen by lot, and referendums on the broad lines, for instance on the share of labour that flows into communal tasks. That is more than Beer had to say on the question of the setpoint.

Why Labour Time Is a Poor Controlled System

The proposal is well thought out, and yet it founders on its unit of account. From the point of view of control engineering, labour time has four weaknesses.

It is sluggish. What counts as socially necessary labour time has to be surveyed, for millions of products, in thousands of firms. By the time the figures are in, the technology has changed. The sensor reports what was yesterday.

It can be manipulated. A firm whose product is valued by labour time has an interest in making its effort appear high, or in relabelling its product so that it falls into a more favourable group. That is the nail factory of Chapter 1 in a new form.

It blurs performance. What is valued is not the labour time a particular firm needs, but the average. Whoever works more slowly is carried along by the average. Whoever is faster does not see their lead in their account. Today's economy knows the same pattern from companies that survive only on cheap credit: the selection that lets weak firms drop out is suspended.

It is blind to everything that is not labour. Here lies the deepest objection, and it is the same as the one against price in Chapter 5. Land, water, energy and the absorptive capacity of the atmosphere are scarce, but cannot be measured in hours of work. A litre of groundwater pumped up in an hour costs an hour, regardless of whether the well then runs dry. Cockshott calculates

everything in labour time, just as Hayek calculates everything in money. Both make the same mistake with different numbers.

What Remains

Despite these objections, the book provides three things that the design in Part Three needs.

The first is the proof that the computational task can be solved. Mises' claim that one cannot calculate without a market dates from a time without computers. Today the question is no longer whether one can calculate, but with what.

The second is the idea of a money that cannot work. The labour credits are cancelled on purchase and cannot be lent. This breaks the positive feedback loop in which property becomes more property. One need not adopt labour time as the unit of account to adopt this thought: that the properties of money itself determine which loops are possible in an economy.

The third is the seriousness with which Cockshott and Cottrell ask the question of the setpoint. Bodies chosen by lot and referendums on the broad lines are a proposal that goes beyond Beer and is taken up again in the chapter "Who Regulates the Regulator?".

The next chapter gives the floor to the critics who consider every form of cybernetic steering dangerous.

If money that cannot be lent bears no interest: how many of the positive feedback loops of Chapter 4 would still be running?

Chapter 10: The Critics – Tiqqun and Overwijk

The Warning

In 2001 a text appeared in Paris without an author's name. It came from a group calling itself Tiqqun and was titled "The Cybernetic Hypothesis". Its thesis was radical: cybernetics was not one science among others but the new principle of governing. It had replaced liberalism as the guiding idea. Its goal was a rationalised, stable and thus harmless society, in which every human being was merely a link in feedback loops that constantly measured and readjusted them.

Tiqqun recommended not better regulation but sand in the works: noise instead of signal, slowness instead of speed, opacity instead of visibility. Whoever evades measurement evades steering.

For a book that sets out to design an economy with built-in feedback, that is a hard objection. It says: what you are building is the apparatus you ought to be warning against.

An Objection from Within

More cautiously, but in the same direction, the philosopher Jan Overwijk argues in his 2025 book "Cybernetic Capitalism". He combines Niklas Luhmann's systems theory with the critical theory of the Frankfurt School. His starting point is an observation that fits Chapter 1 of this book: Hayek already described the market as an information system that processes signals. Today's capitalism is accordingly already cybernetic, only without calling itself so.

The book's subtitle speaks of the incommunicable. What is meant is whatever cannot be translated into the signals of such a system

and therefore does not appear. At its core this is the same finding as in Chapter 5: a sensor that measures only what is sold reports everything else as zero. Overwijk, however, draws a different conclusion from it. The answer, he says, is not a better sensor but the insight that every information system excludes something.

What the Critics See

The warning is not plucked out of thin air. The present supplies illustrations enough.

Whoever drives for a delivery service is directed by an app that measures every minute, sets routes and allocates fewer jobs if the work is too slow. No supervisor shouts; the control loop sees to that. In China, authorities and companies rate the behaviour of citizens and firms with points systems. They are less uniform than Western reports often claim, but they show the direction. And in the chapter on Beer it has already been said that the same telex machines that secured supplies in Chile could have served surveillance in the hands of another government.

The common pattern is clear. A control loop becomes dangerous when it measures people instead of conditions, when sensor and setpoint lie in the same hand, and when the person measured does not know what is being measured and for what purpose. Then the system regulates not the economy but the people, and towards a setpoint they did not choose.

What They Overlook

The critics are right about the danger. They are wrong about the way out.

Noise, slowness and opacity are not an order. A society without feedback is not freer; it is blind. Chapter 1 showed where that

leads: to the nail factory that produces for no one, or to the crossroads without rules, where the strongest drives through. Where nobody measures whether a river is being poisoned, it is not the individual who profits but the polluter.

Above all, the critics overlook a distinction that control engineering knows precisely: the one between steering and regulating. To steer means to give orders from outside that the system carries out. To regulate means that a system brings itself back to its setpoint. An overseer who watches every movement steers. A thermostat regulates. The delivery app steers people. The roundabout from Chapter 1 regulates traffic without watching a single driver.

The cybernetics that Tiqqun fears is the cybernetics of the overseer. The cybernetics this book is looking for is that of the roundabout: rules built into the design and therefore needing to watch no one.

Four Protective Rules

From the critics this book adopts not their way out but their warning, in the form of four protective rules that every proposal in Part Three must observe.

1. **Measure conditions, not people.** What is measured is soils, rivers, stocks, energy flows and money flows in aggregate, not the behaviour of individuals.
2. **Sensor and setpoint in different hands.** Whoever measures must not decide what is regulated towards. And both happen in public.
3. **A right to opacity.** There are areas of life in which nothing is measured, not out of negligence but on purpose. What Tiqqun recommended as resistance becomes a building element here.

4. **Regulate before steering.** Where a design establishes equilibrium by itself, no centre is built. A centre receives only alarms, and even those only when the units themselves are at a loss, as with Beer.

With this, the four lines have been examined. The next chapter brings them together.

If a roundabout orders traffic without watching a single driver: why should a good economic order need to watch its citizens?

Chapter 11: What Holds Up and What Does Not

Four Parts of a Whole

So far the four lines of Part Two have stood side by side in the literature, as if they had nothing to do with one another. Beer wrote for managers, Vester for ecologists, Cockshott for socialists, Tiqqun for insurgents. They hardly cite one another.

Lay them side by side, and something surprising appears. They contradict one another less than they complement one another. Each supplies a piece that the others lack. Beer supplies the structure, Vester the limits, Cockshott the calculation, the critics the protection. Like a house for which a structural engineer, a building physicist, a cost estimator and a fire-safety assessor have each drawn part of the plan without ever meeting.

The Synopsis

The table shows for each line what it provides, where it fails and what the design adopts.

Line	Provides	Fails because of	Adopted
Beer	Structure of viable systems, self-regulation below, alarm with grace period	No sensor for need and nature, setpoint with the government	The five subsystems at every level, alarms instead of orders
Vester	The limits: eight basic rules, the Paper Computer	Politically neutral, no unit of account, applied only on a small scale	The eight rules as touchstones, the Paper Computer as a method
Cockshott and Cottrell	Proof that calculating without a market is possible; money that cannot work	Labour time as the sole unit of account: sluggish, manipulable, blind to nature	The idea that the properties of money determine the possible loops; bodies chosen by lot
Tiqqun and Overwijk	The warning against a system that regulates people instead of conditions	No way out except noise and opacity	Four protective rules, regulate before steering

What None of the Four Provides

However well the four parts fit together, at three points they are all silent.

The positive feedback of power. None of the four lines has an answer to what was described in Chapter 4: that wealth turns into influence, and influence into rules that increase wealth. Beer presupposes a well-meaning government, Vester a sensible municipality, Cockshott a society without capital, Tiqqun no order at all. How an order protects itself against its own beneficiaries, none of them asks.

The form of ownership. Who owns the firms that are supposed to regulate themselves? Beer worked with nationalised firms, Cockshott with common ownership, Vester left the question open. Each of the four lines thus tacitly assumed a form of ownership instead of comparing the forms. This book will favour no form before it has been examined. From the sole proprietorship through the listed public company, the family business and the cooperative to the state enterprise, each form has its own negative feedback and its own positive feedback. What these are must be measured by the same yardstick for all.

The transition. All four describe a state, not the way there. Chile attempted the leap in two years and was stopped. An order that can only be introduced after an upheaval is either never introduced or introduced under conditions that harm it. How to get from here to there without catastrophe is a design task of its own.

The Brief for the Design

From the diagnosis and the examination follows what Part Three must achieve. It must design an order

- whose units regulate themselves and report upwards only what they cannot solve,
- that keeps Vester's eight rules, above all the dominance of negative feedback and independence from growth,
- that calculates with more than one unit of account and whose money does not turn into more money of its own accord,
- that measures conditions and not people, and whose sensor and setpoint lie in different hands,
- that protects itself against the positive feedback of power by giving privileges a built-in end,
- and that can be introduced step by step, without waiting for a catastrophe.

That is a lot. But it is a list of design conditions, not a creed. Each one can be tested, and each one has models that already work somewhere.

If a structural engineer, a building physicist, a cost estimator and fire safety have each drawn part of the plan: who sits down and draws the house?

Chapter 12: The Basic Principle

Cruise Control

Imagine an engineer who designs a car and announces: brakes, clutch, accelerator pedal, all that clutter for regulating speed – we don't need it. We'll fit an automatic device that accelerates steadily, because the faster we go, the better off all the passengers are. Nobody would listen to this engineer.

Since the eighties, the designers of the economy have done something similar and called it deregulation. In cybernetic terms this means: negative feedback is removed, positive feedback stays. Tax havens take away the effect of balancing between rich and poor. The freeing of the financial markets took away speculation's limiter. Market power that creates more market power was left untouched. A vehicle like that drives well as long as the road is straight. 2008 was a bend.

The usual answer to such bends is: more rules. This chapter shows why that alone does not help and what has to be built instead.

Steering and Regulating

A toilet cistern fills itself after every flush and stops when it is full. Nobody has to turn off the tap. The float measures the water level, and the valve closes as soon as the level is reached. Had someone fitted a timer instead that stops the inflow after thirty seconds, the cistern would overflow when the water pressure is high and stay half empty when it is low. The timer steers. The float regulates. The timer follows a plan, the float a measurement.

This distinction is the core of the whole book. Steering means: someone decides in advance what is to happen and hopes the world will stick to it. Regulating means: the system continuously measures where it stands, compares this with the setpoint and intervenes by itself when the deviation grows.

Laws are mostly steering. A regulation that demands a certain equity ratio fixes a number. Whether the number will be enough in the next crisis, it does not know. That is why a state with many regulations is not, for that reason alone, a well-regulated state. It can be a dense network of timers, each of them built for a situation that is long past. Whoever knits the network tighter

creates bureaucracy, but not stability. Whoever tears it apart creates cruise control.

An example of both is the German debt brake. At its core it is steering: a fixed limit on new borrowing. But it contains a part that regulates: in bad years the state may borrow more, in good years it must borrow less. Where it no longer fitted, in 2025 people helped themselves with exemptions and special funds – that is, with new steering – instead of improving the regulating part. That is the pattern: where regulation is missing, exemptions are made, and every exemption has to be fought for politically.

Building the Reaction in Advance

The basic principle of this draft can be put in one sentence: **Negative feedback is built in, not demanded.** The reaction to a deviation is not decided at the moment of crisis, but laid into the structure beforehand.

That this works is shown by the economy itself, in a place hardly anyone talks about. When the economy slumps, the number of unemployed rises, and unemployment insurance pays out more. At the same time incomes fall, and with progressive income tax the tax burden falls faster than income. Both support demand without any parliament having to convene. Economists call these automatic stabilisers. They are the only negative feedback in economic policy that reliably arrives in time, because nobody has to decide on them.

The stock exchanges have the same thing on a small scale. If the American share index S&P 500 falls by seven per cent in one day, trading is halted for fifteen minutes. The first form of this safeguard arose after the crash of 1987. In March 2020 it was triggered four times within ten days. The pause does not end a

crisis, but it interrupts the panic in which everyone sells because everyone is selling.

How different things look when the reaction is only decided in the moment. In the autumn of 2008, governments decided at weekends on rescue packages worth hundreds of billions, under time pressure, with incomplete figures and with representatives of the banks to be rescued in the room. What came out of this was not wrong because those involved were stupid, but because the situation made it impossible to decide otherwise. Whoever builds only in the crisis builds with whatever material happens to be at hand, and with whichever people happen to be loudest.

The same applies in a field that seems to have nothing to do with economics. In defence, retaliation decided in the moment appears unpredictable and therefore dangerous. Retaliation that is known in advance and built into the structure appears calm, and precisely for that reason it deters. Game theory has shown this with a simple pattern. When the political scientist Robert Axelrod pitted computer programs against each other in 1980 to find out which strategy does best in repeated encounters, the simplest one won: cooperate the first time, then do whatever the other did last. Its strength was not cleverness but predictability. Everyone knew where they stood.

Five Components

Built-in negative feedback needs five components. If one is missing, it does not work, or it works against its purpose.

1. **A public setpoint.** What the system is meant to hold must be known before it is needed. A setpoint that those being regulated set themselves, or only learn about afterwards, is no setpoint.

2. **An independent sensor.** Whoever is measured must not do the measuring. The rating agencies, which before 2008 were paid by the issuers of the securities they rated, were sensors owned by the controlled system.
3. **An automatic actuator.** The reaction must occur without anyone deciding on it case by case. Every case-by-case decision is an opportunity to prevent it.
4. **Short dead time.** Between the deviation and the reaction there must not be so much time that the situation has already reversed again. Otherwise negative feedback turns into the pig cycle from Chapter 2.
5. **A limited intervention.** The actuator acts on flows, prices and rules, not on people. It changes what pays off, but not who someone is allowed to be. That is the boundary drawn in the chapter on Tiqqun, and it applies to every component of this draft.

These five components are the touchstone for all the chapters that follow. The forms of ownership, the kinds of money and the levels are measured against them.

Leniency as a Component

Axelrod's winning strategy has a weakness that only becomes visible when mistakes happen. If one of the two players misreads a gesture of the other and strikes back once, the other strikes back too, then the first again. A single mistake becomes an echo that never stops. Negative feedback has flipped into positive feedback of punishment.

In 1992 Martin Nowak and Karl Sigmund showed what helps against this: a generous variant that does not repay every offence but forgives roughly every third one. It remains predictable

enough to prevent exploitation, and it breaks the echo. In a world with mistakes it is superior to the strict variant.

For the economy this means: an order that punishes every deviation relentlessly is not stable but brittle. German insolvency law has understood this. Since 2020, anyone who can no longer pay their debts is released from the remaining debts after three years, provided they keep to the rules of the procedure. Before that it was six years; earlier still, there was no such release at all. The old harshness never made anyone pay who could not pay. It took people permanently out of economic life, deterred founders and brought creditors nothing. The ancient idea of the jubilee year, in which debts are cancelled, is the same insight in an older language.

Leniency is therefore not the opposite of negative feedback, but part of it. It is simply built in advance in the same way: with fixed time limits, fixed conditions, and without anyone having to grant mercy case by case.

Where the Principle Reaches Its Limits

Three limits must be taken seriously.

The first was described by the British economist Charles Goodhart in 1975: as soon as a measure becomes the target of a rule, it loses its value as a measure. Whoever knows what the sensor is watching adjusts to the sensor and not to the matter itself. Every built-in negative feedback therefore attracts evasive behaviour. Chapter 4 showed where this leads: the loop moves into the next shadow.

The second limit is the situation that nobody foresaw. An automatic device can only react to what it was built for. A

pandemic, a war, a collapse of supply chains lie outside any setpoint that could have been fixed in advance.

Here we do not entirely agree. One of us considers automatic action to be the decisive thing: every back door for human intervention will sooner or later be used by those who profit from the deviation. The other counters that every machine needs an emergency switch, and that people must operate it. Both agree that the emergency switch must be harder to operate than normal operation, and that every use of it must have an end that comes about by itself.

The third limit is the most difficult. There are situations in which negative feedback is wrong, because the system is supposed to grow. That is what the next chapter is about.

If a toilet cistern knows when it is full: why shouldn't an economy be able to know?

Chapter 13: The Switchover Point

When Positive Feedback Is Right

Anyone lighting a fire in a stove wants positive feedback at first. The little flame is to heat the wood, the hot wood is to give more flame, the bigger flame is to heat more wood. Without this self-reinforcing start nothing burns. But once the fire is burning, you close the air flap a little. Now the stove is to hold an even warmth and not set the flat on fire.

The same pattern is found wherever something new comes into being. An embryo grows at first by every cell dividing and every new cell dividing again. At some point the division has to stop, otherwise what results is not a body but a tumour. A colony of

bacteria in a nutrient solution grows faster and faster at first, then more slowly, until it reaches a stable size. Drawn out, this gives the S-curve that turns up in biology, in the spread of technologies and in the history of many companies.

In the economy, too, positive feedback is often exactly the right thing. The more solar modules were built, the cheaper they became, and the cheaper they became, the more were built. Between 2010 and 2020 prices fell by around nine tenths. Without this self-reinforcement there would be no energy transition. A young company needs the same movement: the first customers bring money, the money better products, the better products more customers. Whoever brakes every positive feedback from the very start prevents the new.

The question is therefore not whether positive feedback should be allowed, but when it must stop. And who decides that.

The Problem of Switching Over

In a nuclear power station the chain reaction is positive feedback: every fission releases neutrons that trigger further fissions. The reactor is built so that the reaction brakes itself when it gets too hot. At Chernobyl it was the other way round. In this type of reactor, under certain conditions, the reaction intensified when the cooling water evaporated. The positive feedback had no built-in switchover point. On 26 April 1986 that point was passed during a test, and nobody could switch over any more.

Financial history is a history of missed switchover points. The tulip bulbs in Holland in 1637, the South Sea Company in England in 1720, the shares of 1929, the internet companies around 2000, the American home loans up to 2008. Each time there was a phase in which self-reinforcement created real value. Each time it went beyond that point, and each time there was no switching over, but

a crash. The American economist Hyman Minsky turned this into the formula that stability breeds instability: the longer things go well, the more debt everyone dares to take on, until the debt itself is the risk.

At present a new merry-go-round is turning. The makers of computer chips invest in artificial intelligence companies, which use the money to buy computer chips from the same makers. The valuations justify the investments, the investments drive the sales, the sales justify the valuations. Whether there is enough real benefit behind it remains to be seen. Nowhere is a switchover point built in.

Why does nobody switch over in time? Because the decision lies with those who profit from the positive feedback. Whoever is winning right now sees no reason to stop. The government is pleased about tax revenue, the voters about rising wealth, the supervisors do not want to be seen as spoilsports. Chapter 4 showed how the holding force of a rule grows over time. With the switchover point it is the same: the longer the positive feedback runs, the more people depend on it, and the bigger the shock that ends it has to be.

From this follows the first rule for the switchover point: **It must not be set by those who benefit, and not after the fact.** There are three ways to build this.

First Way: a Built-in End

The patent is the oldest and most successful form. Whoever discloses an invention receives, for twenty years, the right to exclude others from using it. During this time the invention is meant to bring in money that pays for its development and encourages further inventions. That is intended positive feedback.

After twenty years it ends, without application, without vote, without exception. The invention belongs to everyone.

How important the fixed end is can be seen from copyright, which has the same purpose but did not keep a fixed end. In the USA it originally gave protection for fourteen years, renewable for another fourteen. Since then the term has been extended again and again, most recently in 1998 to seventy years after the author's death, and for works made by companies to ninety-five years. Each extension came shortly before valuable rights would have expired, and the holders of those rights campaigned for it. An end that the beneficiaries can postpone is no end.

The built-in end is suitable wherever the right duration is roughly known: for funding programmes, tax advantages, protective rights, special permits. It is the application of the symmetry of changeability to positive feedback. It is not the end that has to be decided, but the extension – and it has to be justified afresh and passed with the same majority as the first time.

Second Way: a Threshold on a Measurable Quantity

Some bacteria glow, but only when there are many of them. Each one constantly releases a signalling substance. As long as there are few, the concentration stays low and nothing happens. Once the colony becomes dense enough, the signalling substance crosses a threshold, and all of them switch their behaviour at the same time. Biology calls this quorum sensing. No bacterium decides. The threshold decides.

The economy knows such thresholds but rarely uses them for switching over. German competition law presumes a dominant market position when a company holds forty per cent of a market. Above this threshold stricter rules apply. Banking supervision requires banks to hold more equity the bigger and the

more interconnected they are. Both are a start: growth is not forbidden, but beyond a certain size it costs more.

A particularly good example comes from the promotion of solar energy. From 2012 onwards, the feed-in tariff that the Renewable Energy Sources Act paid for electricity from new installations fell the faster, the more installations had been built in the preceding months. This was called the breathing cap. The support, so to speak, switched itself off as soon as it had fulfilled its purpose. No parliament had to argue about whether the industry had now grown up.

The advantage of the threshold over the fixed time limit: it follows what actually happens and not an estimate. Its disadvantage: every threshold invites people to stay just below it. In France, companies with fifty or more employees face numerous additional obligations. A study from 2016 found a conspicuously large number of companies with exactly forty-nine. A hard threshold creates an edge where the evaders gather. Sliding transitions are better: the obligations rise gradually, as with the breathing cap, so that there is no point at which staying small pays off particularly well.

Third Way: the Decision in Advance

Sometimes there is neither a reliable time limit nor a measurable threshold. Then a human decision remains, but it can be built so that it is not made in the heat of the moment.

Medicine has developed a form for this that is hardly known in the economy. Before a large trial of a new drug begins, it is laid down under which conditions it will be stopped early: because the drug clearly works, because it clearly does harm, or because it obviously achieves nothing. An independent board, which is neither paid by the manufacturer nor runs the trial, checks the

interim results against these criteria fixed in advance. The manufacturer cannot let the trial continue because it hopes for a better result, and it cannot stop it because it does not like the interim figures.

Carried over, this means: whoever sets a positive feedback in motion – a funding programme, an industrial policy, a special zone – lays down publicly and in advance how success will be recognised and what happens if it comes or fails to come. The checking is done by a body that does not profit from the positive feedback. What such a body could look like and who can control it is the question of the chapter “Who Regulates the Regulator?”.

The Counter-Test

In 1841 the German economist Friedrich List set out why a country whose industry is still young may protect it with tariffs. His infant-industry tariffs were to be moderate and temporary: as soon as domestic industry could keep up with foreign industry, they were to fall. That was a switchover point from positive feedback to competition, cleanly justified in theory.

In practice the point was rarely reached. The USA raised its industrial tariffs in the 1860s and kept them at a high level until 1913, long after its industry had become the largest in the world. The German Empire introduced protective tariffs in 1879, which soon served the protection of landowners and heavy industry more than the education of young industries. Who decided whether industry had grown up? Industry itself, through its associations and its influence on parliaments. It always found itself still too young.

There is a counter-example. In the 1960s and 1970s South Korea also promoted its industry massively, with cheap loans and protection from imports. But the support was tied to a

measurable condition: the companies had to export and hold their own on the world market. Whoever missed the targets lost the support. The sensor was the world market, and the supported companies could not bribe it. This tie to performance was not the only thing that made South Korea's rise possible, and it was enforced by a government that was not under democratic control. But it shows that the switchover point can be built.

Hysteresis on Purpose

A heating thermostat does not switch the heating on and off at exactly twenty degrees. It switches it on at nineteen and off at twenty-one. If both lay at the same point, it would switch back and forth at every small fluctuation and wear itself out in a short time. Electronics knows the same component under the name Schmitt trigger.

In Chapter 4 hysteresis was an evil: the memory of rules that can no longer be undone. Here it is a tool. A switchover point needs a gap between on and off, otherwise the system flutters. An industry that loses its support on crossing a threshold must not get it back immediately at the next small setback. The difference between good and bad hysteresis lies in who sets the gap: in the good kind, the designer, in advance and visibly to all; in the bad kind, the beneficiaries, after the fact and for themselves.

What Follows from This

For every positive feedback that this draft permits, four conditions apply.

1. **The switchover point is fixed before the positive feedback begins.** Whoever introduces a subsidy, a protection or a privilege must at the same time say when it ends and by what measure.

2. **The beneficiaries do not set it.** Neither when it is fixed nor when it is checked.
3. **Thresholds slide.** Where a measurable quantity determines the switchover point, the consequences rise gradually, so that lingering just below the threshold does not pay.
4. **The end comes by itself.** It is not the end that has to be decided, but the extension.

It remains open whether the biggest positive feedback of all, growth itself, can also be switched over in this way. An economy whose money comes into being as interest-bearing credit depends on growth, because the interest has to be paid out of the surplus. Whether and how this tie can be undone is the question of the chapter on the kinds of money. Before that, the next chapter examines which forms of enterprise bring negative feedback with them of their own accord.

If every fire needs an air flap: who may close the flap on the stove of the economy?

Chapter 14: Forms of Ownership in the Control Loop

No Ownership Is Neutral

Whoever argues about economic orders usually argues about ownership. Some consider private property the condition of freedom, others common property the condition of justice. Both camps have their favourite form and examine their own less strictly than that of the other side.

This chapter does something different. It treats every form of ownership as a component of a control loop. Every form determines where profits flow, who bears losses and who decides. In doing so it determines which negative feedback operates in a company and which positive feedback can arise. No form is neutral. But none is good or bad from the outset. Whether a form holds up depends on what it is used for.

The examination follows a simple rule: every form is measured by the same yardstick, and each gets its best example and its worst failure.

Five Questions for Every Form

1. **Who is liable?** Do the consequences of a decision return to the one who makes it? That is Eucken's principle from Chapter 1: whoever has the benefit must also bear the damage.
2. **Where do the surpluses flow?** Do they stay in the company, go to owners, to members, to a charitable purpose or to the public?
3. **Who controls?** Who sets the company's setpoint, and who can control the controllers?
4. **What time horizon?** Does the form think in quarters, in years or in generations?
5. **What positive feedback threatens?** Where can possession turn into more possession, size into more power, influence into a rule that protects the influence?

The Forms in Detail

The sole trader and the partnership. The baker, the master painter, the law firm. Here negative feedback is at its strongest, because the owner is liable with everything they have. Whoever

runs their business recklessly loses their house. That keeps people cautious and links decision and consequence more tightly than any other form. The weakness is the flip side: hardly anyone takes on great risks if it means staking their family. A power station or a new drug does not come into being this way.

The GmbH (limited liability company). Since 1892 it has made it possible to let a company fail without the entrepreneur going down with it. That has made countless start-ups possible and is, as described in "The Construction of Mercy", a form of built-in leniency. Its positive feedback lies in the same property: whoever has paid out profits and left losses in the company lets the creditors pay when things go wrong – often suppliers and tradespeople who themselves do not have limited liability.

The listed public company. Since the Dutch East India Company of 1602, it has gathered capital from many for ventures that nobody could carry alone: railways, chemical works, semiconductor plants. Without it, much of the technology taken for granted today would not exist. Its weaknesses are just as well known. Owners and management drift apart, the shareholders change constantly, and the rhythm of quarterly reports shortens the view. When companies put their profits into buying back their own shares to lift the price instead of into research, the price rises and the company ages.

The family business. Many of the world market leaders of the German Mittelstand belong to families who think in generations. Whoever wants to pass the company on to their grandchildren does not sacrifice it for one good quarter. That is a long time horizon, of a kind hardly any other form knows. The weaknesses lie in succession, in disputes among heirs and in the temptation to secure one's own position politically. In December 2014 the Federal Constitutional Court declared the far-reaching

exemptions for business assets in inheritance tax partly unconstitutional. Even a good form can overstretch its privileges.

The foundation-owned company. Carl Zeiss has belonged since 1889 to a foundation established by Ernst Abbe. At Bosch, the charitable Robert Bosch Stiftung holds around 94 per cent of the capital, while the voting rights lie largely with a separate trust company. Profits do not flow to private owners, but into the company and into charitable purposes. That protects against the sell-off and against the short view. The weakness is control: who controls the trustees? Boards that choose their own successors can become a law unto themselves. And foundations are also used to remove wealth from inheritance tax.

The cooperative. Friedrich Wilhelm Raiffeisen and Hermann Schulze-Delitzsch founded the first credit cooperatives in the middle of the nineteenth century so that farmers and craftsmen could help themselves. Today the Volksbanken, purchasing groups in food retailing and housing cooperatives are organised this way. Owners and users are the same people; every member has one vote, regardless of the size of their share. That breaks the positive feedback in which more capital means more power. The weaknesses show with size: decisions become slow, the members drift away from the management. The household appliance maker Fagor, part of the large Basque cooperative group Mondragón, went insolvent in 2013. Common ownership does not protect against bad markets.

The company with bound assets. For years a new legal form has been discussed in Germany in which profits may never leave the company and the shareholders cannot sell the assets. In March 2026 the Ministries of Justice and Finance presented a framework concept. Internationally, the outdoor manufacturer Patagonia chose a similar path in 2022. The form rules out the

positive feedback of the sell-off, because there is nobody who could sell. Even its advocates name its weaknesses: fresh equity is hard to come by if investors never share in the profits. And the question of who controls the trustees arises here just as with the foundation.

The municipal and the state-owned company. Municipal utilities, waterworks, transport companies, the railway. They can plan in decades and take on tasks that never pay for themselves but have to be done. For networks that make sense to build only once, such as water mains, they avoid private monopolies. Their weakness is politics. The owner changes its goals with every election, investments are postponed when the budget is tight, and liability blurs because in the end the taxpayer always steps in. The state of many railway lines and bridges in Germany shows how long a public owner can run a network down.

The Forms at a Glance

Form	Who is liable	Where surpluses flow	Time horizon	Strongest negative feedback	Greatest positive feedback
Sole trader, partnership	Owner, with everything	Owner	One working life	Full personal liability	Small; but hardly any great risks
GmbH	Company, up to the contribution	Shareholders	Open	Fresh start after failure	Losses left with the creditors

Form	Who is liable	Where surpluses flow	Time horizon	Strongest negative feedback	Greatest positive feedback
Listed public company	Company	Shareholders	Quarter to years	Capital market punishes mismanagement	Short view, share buybacks, size becomes influence
Family business	Depends on legal form	Family and company	Generations	Responsibility towards the heirs	Succession disputes, political securing of privileges
Foundation-owned company	Company	Company and charitable purpose	Unlimited	No sell-off	Boards without control, tax avoidance schemes
Cooperative	Company; members' additional contributions, depending on statutes	Members and company	Long	One vote per member	Inertia at large size

Form	Who is liable	Where surpluses flow	Time horizon	Strongest negative feedback	Greatest positive feedback
Bound assets	Company	Only the company	Unlimited	No sell-off, no distribution	Lack of capital, control of the trustees
Municipal, state-owned	In the end the taxpayer	Public sector	Long, but election-dependent	Public services without pressure for returns	Political interference, blurred liability

Form Follows Function

The table shows no winner. Every form has a point at which it regulates better than the others, and one at which it can tip over. What follows from this is not a creed but a design rule, corresponding to Vester's third basic rule: form follows function.

For a network that makes sense to build only once – water, electricity, rail – forms that are not driven by returns fit: municipal, cooperative, bound. For a great venture with an uncertain outcome – a new drug, a new technology – forms are needed that gather a lot of capital and limit losses: the public company, the GmbH. For the workshop, the practice, the shop, the full liability of the individual is a good negative feedback. For an

industrial company meant to last for generations, family, foundation and bound-asset companies have proved themselves.

An economic order with negative feedback therefore needs not one form of ownership but many, and rules that apply to all. Three such rules emerge from the chapters so far. First, no form may be able to shift its losses onto others without paying a price for it. Second, a form's privileges, whether fiscal or legal, expire after a fixed period unless they are justified afresh. Third, switching between forms must be possible, so that a company can choose the form that suits its task and not the one that is most favourable for tax purposes.

If every form of ownership is strong at a different point: why should an economy content itself with a single one?

Chapter 15: The Kinds of Money

The envelopes in the drawer

At the start of each month, many families used to divide the wages into envelopes: one for the rent, one for food, one for clothes, one for the holiday. The money in every envelope was the same. But the envelopes stopped the holiday money from vanishing at the supermarket and the rent from running short at the end of the month. Nobody thought of this as a restriction on freedom. It was a way of holding one's own intentions firm against the temptation of the moment.

Companies do the same with reserves, the state with earmarked budget funds, savers with a building society savings contract. Anyone who manages money seriously sorts it. Only the monetary order itself knows no kinds. A euro is a euro, whether it pays for a pretzel, builds a factory or buys a share on borrowed money.

One lever for everything

This has a consequence that any control engineer recognises at once: the economy has many variables that ought to be regulated, but essentially only one actuator, the central bank's interest rate. It acts on everything at the same time.

When prices rose in 2022, the European Central Bank raised the interest rate from zero to four per cent in little more than a year. That was meant to slow inflation, and it did. At the same time it hit housebuilding: in Germany, around a quarter fewer new homes were approved in 2023 than in the year before, in the middle of a housing shortage. The interest rate cannot tell the difference between a loan for speculation and a loan for a block of flats. It is like a heating system with a single knob for the whole house: if the living room gets too warm, the children's room freezes.

Then there is the way money comes into being. It is not printed by the central bank and then lent out; for the most part it is created by the commercial banks when they grant credit. The Bank of England said so publicly in 2014, the Deutsche Bundesbank in 2017. Whether the loan finances a machine that produces more, or the purchase of an office building whose price rises further as a result, makes no difference to how the money is created. In the second case, exactly the positive feedback described in Chapter 4 arises: credit drives prices, prices raise the collateral, collateral allows more credit.

Same value, different purpose

The proposal of this chapter is this: money is divided into a few kinds. All kinds have the same value; a euro remains a euro. They differ only in what they may be spent on. Steering is done not

through different interest rates, but through the quantity of each kind and through the doors between them.

That the kinds are worth the same is no minor matter. As soon as one kind can be had more cheaply than another, say because investment credit is cheaper than consumer credit, it pays to obtain the cheap one and use it as the expensive one. Shell companies, relabelling and black markets are the familiar results. East Germany experienced this with its Forum cheques, with which Western goods could be bought in the Intershop: they were exchanged on the street for East German marks, at a rate that no authority had set.

At the core there are four kinds, plus a fifth that remains open.

1. **Consumption money.** It is the euro as it is today, as cash and in the account, without any tie. Anyone who goes on holiday, orders online or shops at the market notices nothing of the order of kinds. That is deliberate: the greater part of life stays free and unobserved.
2. **Investment money.** It may be spent only on means of production: machines, buildings, plant, research. It is the kind used by enterprises, whatever their legal form.
3. **Subsidy money.** It flows into tasks that the market does not carry: education, basic research, infrastructure, care. It is tied to the purpose for which it was granted.
4. **Resource-bound quota.** Every citizen receives an equal amount each year for goods that put particular strain on nature, such as air travel. Anyone who wants more has to obtain quota from others.
5. **Democracy money.** A small budget per citizen, which he can allocate to political projects, a citizens' initiative, a project in his municipality's budget. Whether it has to be a

kind of its own is open; it belongs rather in the chapter “Who Regulates the Regulator?”.

The non-return valve

Because consumption money is the free euro, it is also the system’s exit. Whatever has once become euros can go anywhere. Steering must therefore sit at the doors that lead into the euro.

The component for this is the non-return valve, as found in every water pipe: water flows in one direction, not back. Euros can become investment money at any time; anyone who wants to invest should be able to. Investment money turns back into euros by one route only: it is spent on a means of production, and the supplier receives euros. The tie therefore applies to the one who spends, not to the one who receives. Meal vouchers already work this way: the employee can spend them only in a restaurant, and the restaurant gets ordinary money for them.

So the sensor sits at the till. With every payment made in investment money, a check is made whether the purchase lies within the basket of permitted uses. For a machine tool that is simple; for a programmer’s computer it is difficult; for the purchase of an existing flat it is one of the hardest boundary questions in the whole model: is that an investment, or merely a change of owner?

Credit only for the new

The strongest lever lies not with the kinds themselves, but with the creation of money. If banks may create credit for enterprises only as investment money, then new money arises only where something new is built. Anyone who wants to buy a share, an office building or an entire corporation must do so with existing

money. It is not forbidden. It just can no longer be done on borrowed money.

With that, the merry-go-round from Chapters 4 and 13 runs out of fuel. The cycle in which credit drives prices and prices secure the next loan is cut at its root, without anyone having to decide on individual deals. That is negative feedback in the sense of Chapter 12: built in, not demanded.

Whether the interest tie of money, and with it the compulsion to grow, can also be resolved in this way is not answered by this chapter. But it does reduce it: if credit arises only for the new, the interest must be paid out of something that has actually been produced in addition, and not out of rising prices for what already exists.

The quota

The resource-bound quota is the most delicate kind, and it is honest to say so. It is a form of rationing. The difference from the ration cards of the war and post-war years is that prices remain free and only the total quantity is limited, and that everyone receives the same amount.

This kind shows that equal value on paper is not yet equal value in the hand. Someone who does not fly has no use for his flight quota. To him it is worth less than a euro, to the frequent flyer more. In Brazil, where many employees receive part of their income as meal vouchers, these vouchers are widely sold below face value, although that is forbidden. The further a tie misses actual need, the larger the discount and the more certain the black market.

From this follows a rule for all bound kinds: they must be cut so that almost every holder needs them for their purpose anyway.

With an enterprise's investment money that is the case. With the flight quota it is not. Here there are two paths. One forbids passing it on and accepts that unused quota lapses. The other allows open trading: whoever does not fly sells his quota to whoever flies a lot, and receives euros for it. Then money flows from the frequent flyers to the frugal, who are mostly also the poorer. Around 2008 Britain had such a personal carbon quota examined by its environment ministry and set it aside as ahead of its time. Which path is better, this chapter does not decide.

Wages

Whether wages should be a kind of their own is an open question between us.

One of us thinks it right that wages and consumption money be identical. Wages are the core of the individual's economic freedom. In the nineteenth century, workers in many places were paid in tokens that were valid only in the factory owner's shop; this truck system had to be banned by law. Any tie on wages recalls it. What a separate wage kind is supposed to achieve, such as the division into spending and saving, is already achieved today by social security contributions and company pensions.

The other sees in a separate wage kind a sensor that does not otherwise exist: it would measure directly how much of the value added reaches labour and how much reaches capital, and it would offer an adjusting screw at precisely this point. Whether this gain in measurement outweighs the loss of simplicity and freedom has not yet been decided.

Where the model is vulnerable

The fungibility of money. An enterprise that receives investment money for a machine it would have bought anyway

saves euros which it can now use freely. The tie is then merely a label. How well-meant earmarking fizzles out in this way is shown by the special infrastructure fund of 2025: the ifo Institute and the Federal Court of Audit criticised the fact that the earmarked money partly replaced investments that had previously been paid from the normal budget. The tie works reliably only where the bound kind is the only route, as with credit for the new.

The boundary questions. Every basket of permitted uses needs boundaries, and at every boundary dispute and administration arise. The baskets must therefore stay coarse, be publicly defined and, following Chapter 13, have a built-in end. A basket that is not justified anew expires.

Observation. Bound kinds can only be run digitally, and digital payments are visible. That consumption money remains the free euro limits this visibility to enterprises, public budgets and the quota. For enterprises this is reasonable; they keep books anyway. For the individual citizen's quota it is a real limit in the sense of the chapter on Tiqqun.

Abroad. Investment money cannot flee abroad, because outside the system it counts for nothing. The euro can, as today. The only new thing would be the temptation to take out credit where there are no kinds. That is why the level at which such an order can begin is at least that of the euro area, or a pilot zone with clear external borders. The chapters on levels and on the transition deal with this.

If families put their money in envelopes to protect their own intentions: why should a society not be allowed to?

Chapter 16: Signals beyond Price

What this book does not decide

Every regulation needs a setpoint. But behind every setpoint stands a question that no control engineering can answer: in which direction does humanity want to go? More prosperity for a few or enough for all, conquest of space or care of the Earth, more work or more free time, growth of things or growth of possibilities. This question calls for an examination of the whole situation with every conceivable path. It is a subject of its own, and it belongs to people, not to designers.

This book does not answer it. It asks something else: how must an economy be built so that it can follow any direction that is democratically chosen, and so that it can change direction without breaking apart?

The difference can be shown with a ship. Where it sails is decided by the crew. Whether it can turn at all is decided by the shipbuilder. A fully laden supertanker needs several kilometres to stop. Today's economy is like a tanker that also has a locked rudder: because it cannot exist without growth and growing debt, it sails straight ahead, whatever the crew decides. Every other direction appears as a crisis.

Two limits therefore hold regardless of any chosen direction. The carrying capacity of nature, for without it there is no direction any more. And no system may steer the people it is meant to serve, for then it is no longer the crew that chooses. Everything else is open. What this chapter provides are the instruments on the bridge: the sensors that show where the ship is, whatever its course.

A dashboard with a single gauge

Imagine a car whose dashboard has only a speedometer. No fuel gauge, no coolant temperature, no warning light for oil pressure. The driver knows exactly how fast he is going. That the engine is overheating, he finds out when he comes to a standstill.

That is how the economy is instrumented. Its most important sensor is price, its most important overall gauge gross domestic product, the sum of all prices. The man who built this accounting system in the United States of the 1930s, Simon Kuznets, warned as early as 1934 against inferring a country's welfare from it. Robert Kennedy said in 1968 that gross national product measures everything except that which makes life worthwhile.

The gauge is not only incomplete; sometimes it shows the wrong thing. A road accident raises gross domestic product: tow truck, hospital, garage, new car. A forest that is felled appears as income, its disappearance nowhere. A mother who looks after her child herself counts for nothing; paid childcare counts. Why price fails at such points was shown in Chapter 5. Here the question is what must stand beside it.

Three kinds of sensor

The first kind measures nature, and in its own units. Tonnes of carbon dioxide, hectares of sealed soil, cubic metres of groundwater, number of species in an area. These quantities cannot sensibly be converted into euros, because their value depends on how close one is to a limit. The thousandth hectare of forest is worth little as long as there are a million, and priceless when it is the last.

The second kind measures how people are faring. How many have a home they can afford? How long do the sick wait for treatment?

How much free time is left after work and travel? How many children leave school without a qualification? Such gauges already exist; they are just not in the cockpit. In 2019 New Zealand aligned its national budget with such quantities for the first time. In 2016 the German federal government presented a report with more than forty indicators under the title “Gut leben in Deutschland” (Living well in Germany). And for years the National Welfare Index has calculated how different Germany’s development looks when environmental damage and unpaid work are included. What has been missing so far is not the knowledge but the connection: none of these gauges is connected to an actuator.

The third kind measures the systems themselves. How long does an approval take? Where do applications get stuck? Which regulation is circumvented most often? These are the questions that are taken for granted in every factory and that an administration often does not even ask.

Quantities where limits are hard

In 1974 the American economist Martin Weitzman established a simple rule. If damage grows slowly and evenly, a price is the better tool: it leaves those involved the choice of where to save. If damage rises suddenly and steeply beyond a threshold, a quantity limit is better: then it is more important to keep reliably below the threshold than to know exactly what it costs.

The European Union has combined the two. In emissions trading it fixes the quantity of carbon dioxide permitted and lets the price of the allowances form on the market. When too many allowances were in circulation and the price stayed ineffectively low for years, it built in a Market Stability Reserve in 2019. It automatically takes allowances out of the market when the

surplus exceeds a threshold. Since then the price has worked. That is built-in negative feedback in the sense of Chapter 12: an independent sensor, a public threshold, a self-acting actuator.

For the design this means: where nature sets hard limits, a quantity regulates, and the price forms freely within that quantity. The resource-bound quota from the previous chapter is the same idea at the level of the individual person.

The duty to report back

A lift button lights up when the call has been received. This small piece of feedback is the reason nobody presses five times or takes the stairs. A machine that is waiting for a person must tell him that it is waiting, and for what.

The same applies to institutions, and it is missing almost everywhere. Anyone who submits a planning application often does not know for months where it is. German administrative law has a negative feedback for this, the deemed approval (Genehmigungsfiktion): if the authority does not reply within a fixed period, the application counts as approved. Silence becomes a yes. Delay thereby acquires a price for the one who delays, and no longer only for the one who waits. So far this rule is applied only where a law expressly prescribes it, and that is rare.

The design turns this into a principle: every system, whether authority, bank or platform, must be able to say at any time where a case stands, what it is waiting for and until when. And where it misses a deadline, the consequence falls on the system, not on the person.

Who measures

A sensor is only as good as its independence. Until 2009, Greece reported false figures on its budget deficit to Brussels for years. When the true figures became known, the deficit was more than twice as high as stated, and the euro crisis began. The problem was not a lack of rules, but a sensor that belonged to the one it was supposed to measure.

Three requirements follow. The bodies that measure are separate from those being measured, and from those who regulate. Their methods and raw data are public, so that anyone can check the calculation. And the measures change from time to time or are supplemented by random samples, because every fixed measure, according to Goodhart, becomes a target over time and loses its value in the process.

Where measuring becomes dangerous

More sensors mean more data, and more data are a temptation. China's experiments with social credit systems show where it leads when a state measures the behaviour of individual people and attaches consequences to it. That is precisely the steering of people that this design rules out.

The line must therefore be drawn strictly: what is measured is flows, stocks and systems, not persons. How much carbon dioxide a region emits, yes; how much a particular citizen emits, only where he himself manages his quota, and visible only to him. How long an office takes over applications, yes; how often a particular clerk takes a break, no. A sensor that points at people turns negative feedback into surveillance.

If nobody would drive a car with a single gauge: why do we drive an entire economy that way?

Chapter 20: A Case Study: the Inventors' Cooperative

Why inventing of all things

In the years around 1940 the American patent attorney Chester Carlson offered a process with which documents could be copied dry. One company after another turned it down, including IBM and General Electric. Who would need a copying machine when there was carbon paper? Only in 1944 did a non-profit research foundation, the Battelle Memorial Institute in Ohio, take up the matter and develop it further. The invention became Xerox, and the returns carried further research at Battelle for decades.

The story shows in miniature what ails the whole economy. The early phase of an invention needs positive feedback: money, time and patience, before anyone knows whether it will pay. Precisely in this phase, capital as it is built today demands full control as the price for its risk. The inventor faces the choice of remaining powerless without capital or being expropriated with it.

That is why inventing is suited as a case study. It is small enough to survey, and it contains all the problems of this book in the tightest space: the necessary positive feedback, the difficult switchover point, the question of the form of ownership, the danger that an institution forgets its purpose. In the book "The Four Columns of Inventing" we designed such an institution: InnoCoop, a cooperative of inventors. This chapter tests it with the tools of the preceding chapters.

The separation

The core of InnoCoop is a single separation: financing and control are taken apart. A foundation holds the capital and the rights.

Control lies with the inventors' cooperative, which appoints the panel that judges the ideas. The principle is: whoever finances does not steer.

This is the same movement as the non-return valve in the chapter on the kinds of money. Money may go in; control may not go along with it. And it is the same separation that steward-ownership demonstrates, examined in the chapter on forms of ownership, only applied one stage earlier: not with the finished enterprise, but with the act of inventing.

In addition there is a closed cycle. Inventive thinking is taught, those trained invent, the panel selects what is to be supported, the inventions are developed up to proof that they work and then licensed, and the proceeds finance the next generation. Inventing is statistical: most approaches come to nothing, the rare hits carry everything. A single inventor cannot withstand these statistics. A community can.

The switchover point

InnoCoop has had the switchover point from Chapter 13 built in from the start. Until it is proven that an invention works, it is supported without having to pay its way. That is protected positive feedback. After that it goes into licensing and must prove itself on the market. And the industrial property rights from which the licences flow have their own fixed end: after twenty years the invention belongs to everyone.

What is open is who establishes when the proof has been delivered. If that is done by the same panel that selected the invention, it is judging its own success. Here what Chapter 13 described for medical studies applies: the criteria for the proof are fixed before the start and checked by a body that has no stake in the invention.

The iron law

Every institution that is meant to fulfil a purpose is at the same time a system that wants to survive. And because the world continually selects for survival and not for fidelity to purpose, over time the goal shifts from the purpose to the preservation of the institution. The panel closes itself off into a guild, the criteria harden, and in the end the inventors' foundation sifts out precisely the awkward invention for whose sake it was founded.

This is not a question of bad people. It is a selection pressure that operates even among the entirely upright. It is the positive feedback of self-preservation, and it threatens every institution this book proposes, not only InnoCoop.

The obvious answer, limiting the term, has been tried and has failed. In their early days the Greens rotated their members of parliament at the halfway point of the legislative term and separated party office from parliamentary mandate. Rotation destroyed experience, and the apparatus came back through the front door. It is not the duration that is the harmful part, but the coupling of office and self-preservation. The chapter "Who Regulates the Regulator?" must take this experience seriously when it speaks about limiting offices.

The fork as negative feedback

InnoCoop takes hold at two points. First, the purpose is made irreversible: the assets are tied to it and cannot be sold or privatised. This part is proven; foundation-owned companies have shown for decades that it holds. But it protects only the letter. It does not stop anyone from keeping the shell and hollowing out the purpose from within.

Second, and this is the genuinely cybernetic part, a sensor is built in that lies outside the institution. The model comes from free software. When Oracle took over the office program OpenOffice and it stagnated, the community of developers split off the work in 2010 and continued it under the name LibreOffice. No court had to rule on betrayal of purpose. The people left, and the original withered. Programmers call this a fork.

Transferred, this means: the community of inventors has the right to refound if the leadership betrays the purpose, and the purpose-bound assets follow the active community, not the apparatus. The sensor is then no longer a text that someone has to interpret, but an observable behaviour: are the inventors leaving or not? That is the difference between a judge ruling on a company's failure and customers who stop buying.

This does not solve the problem. Program code can be copied for free; money cannot. For the assets to follow the fork, a rule is needed for when they follow, and this rule is again a point at which one can take hold in order to capture it. How to build this trigger cleanly is InnoCoop's highest open design task. It is at the same time one of the most important of the whole book, for every institution proposed here faces the same question.

What the case study shows for the whole

Four lessons reach beyond inventing.

1. **Separate financing and control.** Where money automatically brings power with it, the positive feedback of Chapter 4 arises. Separation is the most effective component against it.
2. **Make the purpose irreversible, but do not rely on it.** The legal form holds the letter, not the meaning.

3. **Place the sensor outside.** An institution that checks itself checks its own continued existence. The most reliable sensor is the behaviour of those for whom it exists.
4. **Do not leave the upright on their own.** The afterword of “The Four Columns” sets out the prisoner’s dilemma of the upright inventor: whoever does not externalise costs loses on price to whoever does. The individual cannot resolve it. A structure in which uprightness pays is exactly what this book is trying to build.

Where the example is open to attack

InnoCoop does not yet exist. It is a design whose components have been tested individually, but never together. Its most impressive model is at the same time a cautionary case: Nikolaus Laing showed over decades that an independent inventor can live from inventing and licensing, and in the end failed because he became dependent on outside capital and his work hung on his person.

The start-up capital proposal has a further weakness. The companies that have profited for decades from a process that has become free are supposed to contribute part of their proceeds voluntarily. Voluntariness is not negative feedback. There is no sensor and no actuator, only the hope of insight. And it is the process of one of the two authors of this book that is at stake. That is no reason to keep the proposal quiet, but it is a reason to examine it with particular rigour. A case study that concerns its authors proves nothing as long as it has not been realised.

If a community of programmers saves its purpose by leaving: how do you build an economy in which leaving is possible and the assets go along?

Planned economies failed because nobody could report back. Today's markets fail where their feedback runs the wrong way: bubbles, ratchets, exhausted fishing grounds – and states that have to rescue them in every great crisis.

This book treats the economy as what it is: a web of control loops. It asks where the negative feedback is missing that every lasting system needs, and drafts an order in which it is built in from the start instead of demanded after the fact – like a roundabout, which needs no traffic light and no police officer, only a clever shape and one simple rule.

From Stafford Beer and Frederic Vester to kinds of money, forms of ownership and the switchover point at which growth must turn into balance: the draft of an economy that regulates itself.

Which direction it should take is not for this book to decide. That belongs to the people.

